FT-ITC Analysis
User manual

Processing, fitting, and interpreting isothermal titration calorimetry experiments.

VERSION 1.5.0 · FOR MACOS, WINDOWS, AND LINUX

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CHAPTER 01

FT-ITC Analysis user manual

Start here for a tour of FT-ITC Analysis and the conventions used throughout this manual.

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In isothermal titration calorimetry (ITC), an instrument measures heat as one solution is added to another. FT-ITC Analysis turns those measurements into heat values for individual injections, fits a proposed explanation of the interaction, and helps you inspect and present the result.

This manual is for ITC practitioners who are new to FT-ITC Analysis. It explains how to use the application and provides guidance on scientific interpretation; no programming knowledge is required.

These instructions apply on macOS, Windows, and Linux. The manual treats the supported desktop applications as one product. A **Platform note** appears only when an operating-system or interface difference changes how you complete a task.

VERIFIED:
This edition reflects FT-ITC Analysis 1.5.0 and was verified on the date shown for each page.

What the application does

The workflow has two main questions: **How much heat did each injection produce?** and **What interaction could explain those heats?** Processing answers the first question from a raw power trace (a **thermogram**). Fitting compares the injection heats with the predictions of a chosen model. A close fit is useful evidence, but it does not prove that the model describes the chemistry.

Start with Quick start: import data, check experiment details, process any thermogram, fit a model, review the result, then save and export. Use the task chapters below for setting choices, calculations, and interpretation limits.

Raw input files are not modified. Ordinary desktop analysis runs locally. Optional online interpretation and the browser-based Project Viewer send data only through their respective workflows; review Privacy and online checks before using them.

Product tour

The data list contains loaded experiments and completed Analysis Results. Selecting an experiment exposes four shared task views:

- **Overview** summarizes the experiment and provides access to its details.

- **Process Data** builds and refines baselines, adjusts injection regions, and converts a raw thermogram into integrated heats with estimated uncertainties.
- **Analyze Data** fits a single experiment or multiple experiments.
- **Final Figure** presents the thermogram, heats, fitted curve, residuals, and annotations.

Selecting an Analysis Result opens its saved fits, parameter summary, uncertainty, correlations, and supported advanced analyses. See Workspace for navigation and Results and advanced analyses for interpretation.

Manual conventions

The shortest route to a first result is Quick start, which uses your own compatible data. The relevant task chapter provides setting details and diagnostic guidance when a workflow needs closer inspection.

Manual directory

- Quick start
- Installation, files, and projects
- Workspace
- Processing
- Single-experiment fitting
- Multiple-experiment fitting
- Results and advanced analyses
- Figures and export
- Tools
- Settings and defaults
- Glossary and equations

The manual uses these callouts:

> **NOTE:**
> A detail that affects how the application behaves.

> **CAUTION:**
> A condition that can produce a failed task, invalid result, or misleading output.

INTERPRETATION:
Scientific context for judging output; it is not an automated conclusion made by the application.

RECOMMENDATION:
A defensible working practice rather than a software requirement.

PLATFORM NOTE:
A genuine operating-system or interface difference that changes how a task is completed.

CALCULATION:
A compact relationship used by FT-ITC Analysis or needed to interpret its output.

Interface labels appear in **bold**. A path such as **File > Save As...** means choose the menu and then the command. The names of views and controls are used instead of position-dependent phrases, so the instructions remain usable when the window size or platform changes.

Help and support

Help and Guide opens the topic-based in-app help, with practical guidance followed by scientific explanations and calculations. On native macOS, choose **FT-ITC Analysis Help** from the **Help** menu. **Citation** presents the current paper and versioned software citations and supports copying or exporting BibTeX. **Contact Support...** helps you contact support by email and prepares a diagnostic report containing the application version, operating system, recent activity, and full application log. **Copy Support Report** places that report on the clipboard. Official links include the FT-ITC Analysis website, latest release, project viewer, and software DOI 10.5281/zenodo.14832177.

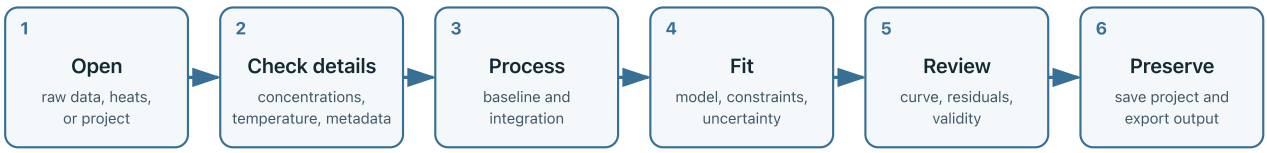
CHAPTER 02

Quick start

Open your own ITC dataset, process it, fit a one-set-of-sites model, review the result, save the project, and export a figure.

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This walkthrough takes a compatible file from measured heats to a fitted result. A **One-Set-Of-Sites** model is a starting example when the binding sites can reasonably be treated alike; choose another model if the chemistry calls for it. The steps show how to use the application, not how to decide which scientific model is true.



The overall analysis workflow. Integrated-heat imports enter after thermogram processing.

Before you begin

Use a compatible raw thermogram, integrated-heats file, Origin project, or `.ftxtc` project; see Supported input formats. Integrated heats and Origin projects without a usable time/power trace skip thermogram processing.

1. Open your data

Launch FT-ITC Analysis and choose **Open File...** on the welcome screen, choose **File > Open...**, or drag compatible files into the application window. Select your file and choose **Open**.

Select the imported experiment in the data list and open **Overview**. See Workspace for selection, Active experiments, and list ordering.

2. Edit experiment details

Open **Details...** for the selected experiment. Concentrations, conditions, comments, and attributes can be edited here. A date read from the data file is trusted and read-only. A filesystem timestamp does not establish the experiment date; entering a replacement marks it as **User provided**. The date source is shown beside the date controls.

Tandem experiments keep their generated merge description under **Tandem merge origin**, separately from editable comments.

> **CAUTION:**
> Apply corrections only when you have an independent experimental basis. Concentration entries influence the calculated concentration ratio and fitted parameters.

3. Process a raw thermogram

If the import contains a thermogram, open **Process Data**.

1. The baseline estimates the signal the instrument would show between injection peaks. If that background drifts smoothly, try **Polynomial**. For a more irregular background, try **Spline**; for changes that differ across parts of the run, try **Segmented**. Keep the default integration settings initially.
2. Inspect whether the baseline represents the signal between injections rather than the peaks.
3. Inspect the start and end of every integration region. A region should include the injection response without extending unnecessarily into baseline noise.

These baseline alternatives are explained in Processing.

> **RECOMMENDATION:**
> Double-click a peak to zoom in, then press **Space** to copy its integration end-point offset to the next injection.

> **INTERPRETATION:**
> A visually smooth baseline is generally desired. As a rough rule of thumb, aim for at least 20% of the data points to be baseline data; this is guidance, not a hard threshold. Apparent jumps in the baseline may indicate insufficient equilibration time between injections. Major spikes may require advanced baseline editing; the **Spline** baseline type provides more flexibility.

4. Fit a model

Open **Analyze Data**, choose **Single experiment**, and select **One-Set-Of-Sites** or a more appropriate model, depending on your system.

1. Review the initial parameter values: affinity describes how strongly the substances bind, enthalpy describes the heat change associated with binding, and stoichiometry describes the binding ratio in the chosen model. Use physically plausible starting values.
2. Choose an optimizer. **Levenberg-Marquardt** is efficient near a suitable solution; **Nelder-Mead** provides a derivative-free alternative when convergence is difficult.
3. Optionally enable **Weight by injection error** when the integration uncertainties are meaningful for the dataset.
4. Choose an uncertainty method. **None** skips parameter uncertainty; **Bootstrap residuals** refits simulated variations of the data; **Leave-one-out** checks how much the fit changes when an observation is omitted; **Profile likelihood** tests how far a parameter can move while the fit remains acceptable.
5. Choose **Run Fit**.

Create analysis result determines whether a usable fit also creates a separate Analysis Result. Enable it to continue through the result workspace in the next step; otherwise, the fitted solution remains attached to the Experiment Data.

> **RECOMMENDATION:**
> More complex models such as the two-sets-of-sites model may require considerable trial and error with starting parameters in order to obtain a good solution.

If the fit fails or reaches a limit, the possible causes are described under Fit availability and non-convergence.

5. Review the result

Inspect the fitted curve together with the residuals (observed heat minus predicted heat). Random-looking small residuals are more reassuring than a repeated pattern the model misses. When **Create analysis result** is enabled, select the resulting **Analysis Result** and check:

- the included experiment and injections;
- fitted parameter values and units;
- convergence status and RMSD;
- whether weighting and uncertainty estimation match your intention;
- parameter uncertainty or confidence intervals, when calculated;
- parameter correlations.

> **INTERPRETATION:**
> Parameter precision does not establish model adequacy. Look for structured residuals, sensitivity to initial values, values at limits, excessive parameter correlation, or disagreement with known stoichiometry and experimental conditions.

6. Save a portable project

Choose **File > Save As...** and save the project as `.ftxtc`. The project contains imported data, experiment details, processing state, fits, results, and available advanced-analysis output. Keep the original instrument files as source records even though the project is portable.

If autosave is enabled, it supplements normal saving; it is not a replacement for a named project file.

7. Export a figure

Return to the experiment and open **Final Figure**. Choose the elements you need, such as the thermogram, integrated heats, fitted curve, residuals, confidence band, or parameter information. Review axis ranges and labels, then use the figure export or print command provided by the view.

Use **Analysis Result Exporter...** when you need numerical result tables rather than a graphic. For injection-level heats, choose **File > Export Integrated Peaks...**. The same output is available through **File > Export Data...** by choosing **Integrated Peaks**.

The figure and table workflows continue in Figures and export.

> **NOTE:**
> An Analysis Result can export a figure using its saved fit for each member experiment. Use **Export Associated Final Figures...** on the result to make those figures without first loading the saved solutions into the Experiment Data.

CHAPTER 03

Installation, files, and projects

Install FT-ITC Analysis, open supported data, save portable projects, and use autosave or recovery safely.

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Install the application

Use the latest FT-ITC Analysis release for current packages and release notes. Verify that the file came from this official release page before accepting an operating-system security prompt. Package options may change between releases. For installation help, contact `support@ft-itc.org`.

macOS

Open the DMG, drag **FT-ITC Analysis** to **Applications**, and eject the disk image. If macOS blocks a verified download, review it in **System Settings > Privacy & Security** and use **Open Anyway** only after confirming its source.

Windows

Run the supplied Windows x64 `.exe` installer and follow the setup prompts. The installer registers `.ftxtc` project associations. If Windows displays an **Unknown publisher** or Microsoft Defender SmartScreen warning, continue only after confirming that the installer came from the official FT-ITC release page.

Linux

Install the supplied `.deb` package matching the system architecture (AMD64 or ARM64, when available) on a compatible Debian-based distribution. Package trust and dependency behavior depend on the channel through which the package was obtained. Review the current release notes for known platform limitations.

> **PLATFORM NOTE:**
> Package installation and file-association prompts are controlled by the operating system. The analysis workflow is otherwise shared; the manual notes the few interface labels that differ between desktop editions.

Supported input formats

Choose the input that matches the stage of your work. A **raw thermogram** contains the instrument's power trace and still needs processing. An **integrated-heats** file already contains a heat value for each injection, so it goes directly to fitting. A **project** can reopen earlier FT-ITC work.

EXTENSION	SOURCE	IMPORTED CONTENT
`.itc`	MicroCal-style raw data	Raw thermogram and injections
`.nitc`	TA Instruments NanoITC native data	Raw thermogram and injections
`.ta`	TA Instruments/NanoAnalyze export	Raw thermogram and injections
`.apj`	PEAQ-ITC project	Raw thermogram and injections from each experiment
`.opj`	Origin project file	Raw thermogram or integrated heats, according to the recognized worksheet
`.dat`	Integrated-heats table	Integrated heats and injections
`.aff`	Integrated-heats table	Integrated heats and injections
`.dh`	Fixed-layout integrated-heats file	Integrated heats, injections, and experiment metadata
`.ftxtc`	Current FT-ITC project	Stored project state

Raw thermogram imports (`.itc` , `.nitc` , `.ta` , and `.apj`) open as unprocessed Experiment Data. Process and fit them in FT-ITC Analysis. A PEAQ-ITC `.apj` import adds every experiment in the file; its existing processing and fits are not imported.

For an Origin `.opj` file, FT-ITC Analysis uses the first recognized ITC worksheet. It imports the raw thermogram when available, or integrated heats otherwise. Origin processing and fits are not imported. The newer `.opju` format is unsupported.

If a MicroCal `.itc` file contains concatenated runs, an import prompt offers the standard **Use MicroCal Concat** calculation or **Use Back-Mixing Compensation**. Use the latter when you have the required mixing information. The imported experiment's **Tandem merge origin** records the number of runs and the calculation applied. See Experiment Merger for details.

Integrated-heat files (`.dat` , `.aff` , and `.dh`) skip thermogram processing. During import, select the heat unit used by the file and supply any requested concentration information. For `.dat` and `.aff` , **Recalculate concentrations and ratios** replaces the file's concentration progression with one calculated from the injections. These two formats do not record the experiment temperature, so the user is prompted to enter it. It can be corrected later in the experiment details. Choose **Use selected action for remaining files** only if those files use the same heat units, temperature, and import choices; leave it unchecked when importing a temperature series. These formats contain no thermogram, so they cannot restore a baseline or processing-derived injection uncertainties.

The first injection is imported but excluded from fitting by default because its heat is often unreliable. You can include it manually; its volume still contributes to later concentration calculations.

A zero cell concentration is retained, for example for Dissociation into buffer; plots then use titrant concentration instead of molar ratio. Review import warnings about concentrations, cell volume, or gaps of 10 s or more in the thermogram, which can affect baseline fitting and integration.

Open files

Choose **File > Open...**, use the welcome-screen action, or drag files into the application. Multiple supported files can be opened together. Files opened into a populated document are added to its existing Data / Results list; this includes `.ftxtc` projects. Clear the current document first if you want to open a project by itself.

If a project file contains experiments, Analysis Results, reports, or saved fits already loaded, choose **Skip Duplicates** to keep the existing objects and add new content, including new results that use existing experiments. Choose **Import Copies** to add separate copies while keeping their references connected to the copied content. **Skip Duplicates** is the default.

Opening one complete `.ftxtc` project into an empty document makes that file the destination for **Save**. Adding files to an existing document keeps its current save destination. If the document has not been named, or you open multiple projects together into an empty document, **Save** asks for a name and location.

> **CAUTION:**
> Appending can create similarly named experiments or results. Confirm the data list and details before fitting or exporting.

Traceability Mode at startup

With Traceability Mode enabled, confirm the operator at launch before recovery or requested files are opened. **Continue** saves the name. **Quit** or closing the confirmation exits without changing it or resolving recovery files.

Save projects

Choose **File > Save** to update the current named project, or **File > Save As...** to choose a new name or location. Use the current `.ftxtc` format for ongoing work.

Save after fitting or processing finishes. Save commands requested while busy are declined; choose the command again afterward. Autosave waits for the operation to finish.

If a save finishes but newer changes remain unsaved, the document stays marked as changed. When this happens while closing, quitting, clearing, or replacing a project, the app keeps the current document open; save again to include those changes.

An `.ftxtc` project preserves the data and metadata needed to continue analysis, including thermograms where imported, concentrations and uncertainties, attributes and comments, injection inclusion, processing state, fit solutions, Analysis Results, and completed derived analyses. The package is portable and does not depend on the original raw-file path for ordinary reopening. Optional external experiment, cell sample/batch, and syringe sample/batch IDs are saved as experiment metadata; each result also retains the operator recorded when it was created or successfully updated.

Save Selected... writes selected project content when you need to share a subset of the project. Confirm the selection before saving and reopen the result if the subset is critical. Selecting Experiment Data saves each selected experiment and its attached solution, if any. If the experiment uses Buffer Subtraction, the reference's heats are stored with it so the correction can be reproduced without the reference experiment. Selecting an Analysis Result saves the result, its member experiments, and the buffer reference experiments they use. It also saves any Analysis Results selected in their **Competitor properties** attributes, with those results' experiments, so the competitor source stays available when the file is opened on its own.

> **RECOMMENDATION:**
> Save a processed version of the project before fitting if you want a reusable starting point. After fitting, save the project again—under a new name if you want to preserve the processed-only version—to retain the fitted solutions and Analysis Results.

Autosave and recovery

Autosave behavior is configured in **Preferences...**. An autosave due during fitting or processing waits and runs when the operation finishes. When recovery data is available after an interrupted session, the application offers a recovery path. Open the recovered document, inspect the data list and recent changes, then save it as a named `.ftxtc` project.

Recovery can restore unaffected experiments, fits, and results. If an experiment cannot be recovered, its fit and dependent results are omitted rather than reconstructed from a different set of members. Review the recovery warning and log, then use **Save As...**; the original file is not repaired in place.

> **CAUTION:**
> Recovery cannot guarantee that every optional result or saved analysis survived. Confirm experiment counts, processing, fits, and result validity before continuing.

Remove and clear content

Removing an experiment or Analysis Result changes only the open document; it does not delete the original raw file. **Remove All Data/Results** clears the current document after confirmation. **Delete All Results** removes all saved Analysis Results while retaining experiment processing and attached fits.

Clear Processing > Selected / Active / All restores processing defaults and removes attached models and fitted solutions for the chosen experiments. Thermogram experiments lose their calculated baselines and integrated heats; imported integrated heats are retained. Raw data, concentrations, injection choices, and attributes—including buffer-subtraction setup—are preserved. Saved Analysis Results remain available; their existing validity checks identify changed experiment inputs.

Clearing multiple experiments requires confirmation. Clear Processing has no Undo and is unavailable while processing or fitting is running. After clearing, process and fit the experiments again as needed.

Saving after removal makes the removal part of the saved project. Use **Save As...** first if you want to preserve the original project version.

Privacy and online checks

Processing, fitting, saving, recovery, export, and printing run locally. Optional automated interpretation in **Analysis Report** sends the selected report evidence, question, and context to an online service when you request generation. See Analysis Report for details.

If **Check for updates and online resources on launch** is enabled, the application checks for release and citation updates; it does not upload experiment data. Disable this setting to prevent launch-time checks; it does not control automated interpretation requests. A failed or disabled check does not prevent local processing, fitting, or saving.

Online data flow and retention

Automated interpretation: Choosing **Generate** sends the selected report evidence and your question or context to FT-ITC's online interpretation service at app.ft-itc.org. The service uses your selected provider, OpenAI or Mistral, to generate the interpretation. Mistral requests use Mistral's EU regional endpoint. This sets where the interpretation is generated, not where all account or operational records are kept. Sent evidence can include experiment and result names, comments, conditions, concentrations, saved fits, uncertainty summaries, diagnostics, and injection data. The selected **Evidence level** limits fitted result detail; Public access uses Results. Supporting experiments contribute summary information at every level, without injection rows or traces. Compressed thermograms for fitted result members are off when the dialog opens and are sent only if you opt in. **Save package** saves the full local evidence and the exact payload for the selected level without sending them. Opening Preferences or the generation dialog can check service availability and account access without sending scientific evidence. A supplied access code is sent to verify access.

Project Viewer: Opening a project in the browser-based Project Viewer uploads the complete `.ftxtc` file, including its embedded data and comments, to app.ft-itc.org for display. This does not submit the project for automated interpretation or send it to an AI provider. The viewer has a 50 MB upload limit. Closing the viewer does not retract an upload or guarantee that temporary processing copies or infrastructure records have been deleted; the full service has no verified end-to-end deletion schedule.

Retention: FT-ITC's interpretation service retains usage metadata that can be linked to requests or accounts, without automatic expiry. These usage records do not contain the complete scientific report evidence or generated interpretation. There is no in-app control to delete service records.

Canceling generation or deleting a local report does not retract information already sent. Data sent to OpenAI or Mistral is subject to that provider's applicable data controls and retention policies; see OpenAI data controls and Mistral regional inference. No zero-retention or fixed-deletion guarantee is made for the complete service flow.

Update safely

Before installing a new application version, save important projects and retain the current installer when reproducibility policy requires it. After updating, open a copy of a representative project, confirm processing and results, and save only when you intend the project to be written by the new version.

CHAPTER 04

Workspace

Navigate the application window and work with Experiment Data and Analysis Results.

Last verified: 2026-09-23

Application window

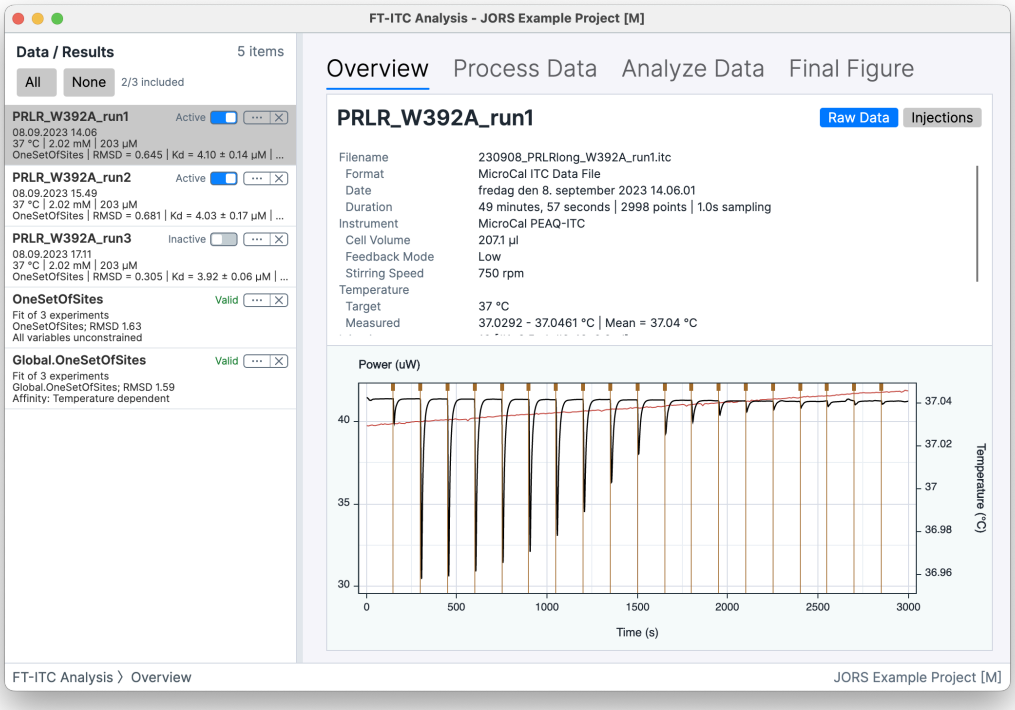
Selected means the item currently shown in the workspace. **Active** means an experiment is included in operations that use a group of experiments. Selecting a row does not automatically make it Active.

The **Data / Results** list is the project navigator. It contains two types of project item: Experiment Data and completed Analysis Results. Selecting an item shows it in the workspace.

Only Experiment Data has an **Active** toggle. **Active** experiments participate in operations that use a group of datasets, such as multiple-experiment fitting, processing propagation, and coordinated export. The toggle becomes available after the experiment has been processed. **Enable All**, **Disable All**, and **Invert Active** change several experiments. The **Sort** commands order the list by name, date, temperature, type, ionic strength, or buffer ionization enthalpy. Multiple-experiment fitting is described in Multiple-experiment fitting, and the processing prerequisite is covered in Processing.

The workflow navigator selects the task; the workspace and controls update for that task. Item actions apply to the selected experiment or result, while the application menu provides project commands, tools, preferences, and help.

Experiment Data



Selecting Experiment Data opens its experiment workflow in the shared workspace.

Experiment Data represents an imported or application-created dataset. Its list entry provides identifying information and a summary of its processing and fitting status.

Experiment workflow

Four task views are available for Experiment Data:

- **Overview** presents the imported data and available experiment information.
- **Process Data** turns a raw thermogram into integrated heats through baseline modeling and injection-region control.
- **Analyze Data** configures and runs single- or multiple-experiment fitting.
- **Final Figure** prepares the thermogram, integrated heats, fit, residuals, and annotations for presentation.

Details and attributes

The **Details...** view contains editable concentrations, comments, and experiment attributes. Use **Details** for the name, date/time, conditions, concentrations, and comments; **Identifiers** for external labels; and **Attributes** for optional metadata. Attributes describe conditions and analysis inputs such as buffer, salt, ionic strength, competitor, or prebound species.

Buffer pH entry requires a finite number from 0 to 14, inclusive.

Use **External experiment ID**, **Cell sample/batch ID**, and **Syringe sample/batch ID** on the **Identifiers** tab of Experiment Details to connect an experiment with your own records. These optional text labels allow duplicates and do not affect processing or fitting. Choose **Apply** to save them. Correcting an ID updates the displayed labels for existing results and newly generated reports; an already generated report preview must be updated to include the correction.

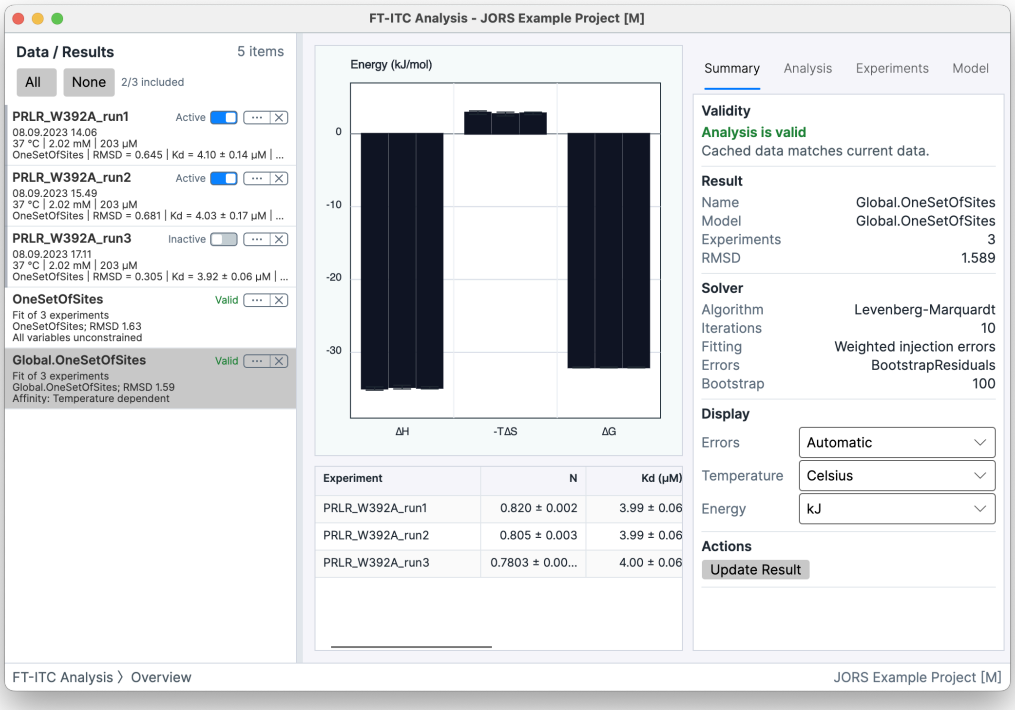
With Traceability Mode and its import-ID prompt enabled, review identifiers after raw-data imports. Select rows to fill blank IDs or replace selected cell/syringe IDs with shared values. **Apply** saves edits; **Skip** or closing retains the imported data without those edits. Project imports do not trigger the review.

Copy Attributes to All copies all attributes from the selected experiment to every other experiment. **Attribute Operations…** copies either one attribute or all attributes to **All other experiments**, **Active experiments**, a **Specific experiment**, or **Experiment names containing…**. The name option targets every other experiment whose name contains the entered text, without regard to capitalization. **Clear Attributes** removes all attributes from the selected experiment after confirmation.

Experiment Data actions

- **Details…** opens the editable experiment information and attributes.
- **Duplicate Data** creates another project item from the selected Experiment Data.
- **Export Selected Data…** exports the selected dataset.
- **Clear Solution** removes the fitted solution currently attached to the Experiment Data.
- **Remove Data** removes the item from the open project without deleting its source file.

Analysis Result



An Analysis Result opens its dedicated result workspace when selected.

An Analysis Result stores a fit for one or more experiments together with its model, constraints, fitting settings, uncertainty output, and member solutions. Its list entry summarizes the stored fit.

Result workspace

Select **Summary** to compare member values, **Fit** to inspect a member's curve and residuals, or **Correlation** to inspect residual-bootstrap parameter relationships. **Temperature**, **Salt**, and **Protonation** require suitable models and metadata.

The inspector organizes result information under **Summary**, **Analysis**, **Experiments**, and **Model**. See Results and advanced analyses for interpretation and prerequisites.

Analysis Result actions

- **Details...** edits the result name and comments.
- In Avalonia, **Copy Result Table** copies the selected result table. On macOS, open **Analysis Result Exporter** and click **Copy to Clipboard**. The exporter also provides controlled table export.
- **Update Result** reruns the stored model and fitting settings using the current member experiments and replaces the result only after a successful fit. For bootstrap uncertainty, choose the stored request or another offered count; the update creates a fresh set of refits.
- **Set Active Experiments** makes the result's member experiments Active.
- **Load Solutions to Experiments** attaches the stored member solutions to their corresponding Experiment Data.
- **Export Associated Final Figures...** exports one figure per member experiment using that result's saved fit. It loads each saved solution only while making its figure, then restores the experiment's previous solution.
- **Remove Result** removes the Analysis Result from the open project.

Separate tools

The **Tools** menu opens separate windows for tasks outside the main experiment and result workflows. These include **Experiment Designer...**, **Buffer Subtraction...**, and **Experiment Merger...**. Their features are described in Tools.

CHAPTER 05

Processing

Choose and edit baselines, adjust integration regions, interpret injection uncertainties, and copy or lock processing settings.

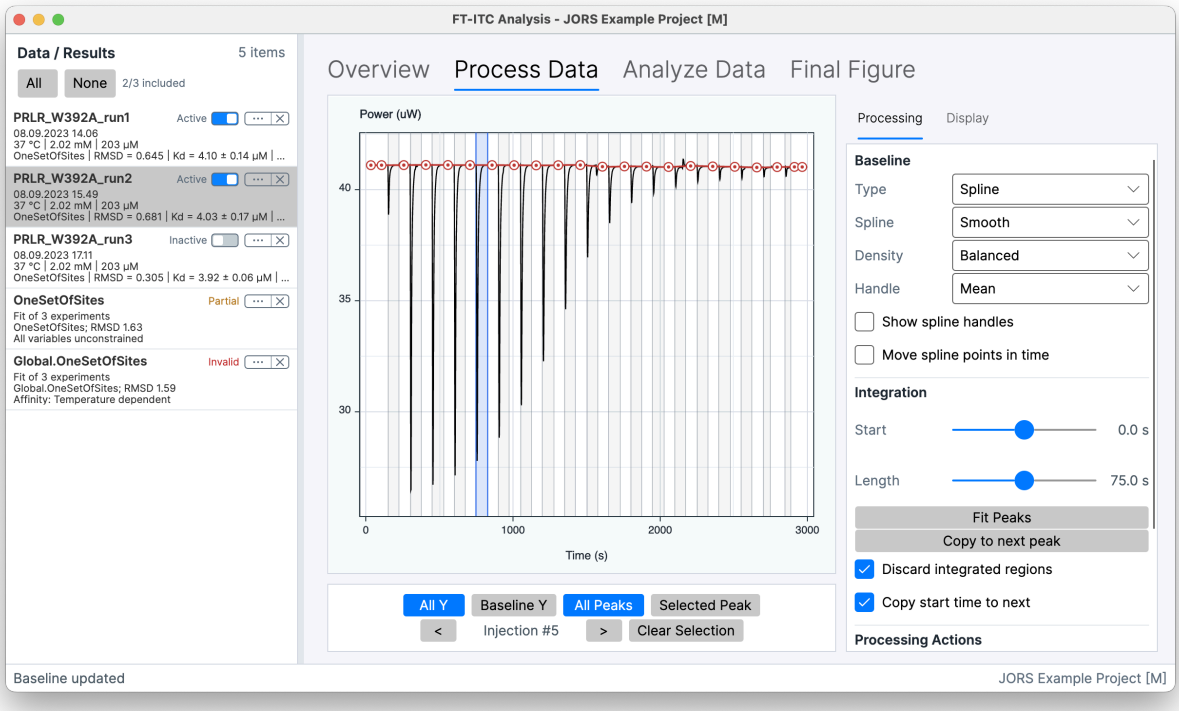
Last verified: 2026-09-23

Open **Process Data** for experiments with a raw thermogram. Choose a baseline, inspect every integration region, then lock processing when it is ready. Integrated-heat inputs skip this stage.

The baseline should follow the signal between injections, and each region should capture the response without unnecessary baseline noise. Use Polynomial or Segmented for suitable automatic baselines and Spline for manual control.

For optional experiment/sample IDs after import, see Details and attributes and General preferences.

Processing workspace



Process Data combines the thermogram with baseline, integration, display, and navigation controls.

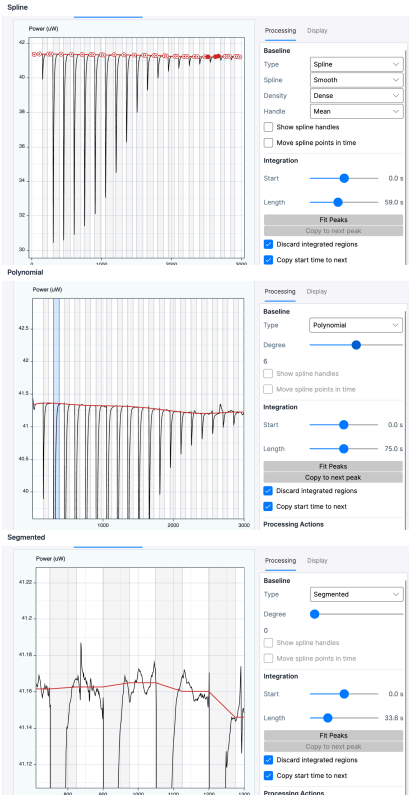
The **Processing** controls configure the baseline and integration regions. The **Display** controls show or hide the baseline, integration regions, corrected data, and cursor information.

With **All injections** selected, **Start** and **Length** apply to every injection. A selected graph region targets one injection, and double-clicking it focuses the graph on that peak. The previous and next controls move through the injections; **Clear Selection** returns to all injections.

The view controls separate horizontal and vertical scaling:

- **All Y** shows the complete power range, while **Baseline Y** emphasizes the baseline region.
- **All Peaks** shows the full injection series, while **Selected Peak** focuses on the current injection.
- Dragging an empty part of the graph zooms into the selected region.

Baseline models and editing



Spline, Polynomial, and Segmented provide different baseline representations for thermogram processing.

Spline

Spline places points in usable baseline regions and interpolates between them. It provides the most direct graphical control.

- **Linear** connects the spline points with straight sections. **Smooth** produces a continuously varying baseline and supports editable slope handles.
- **Sparse**, **Balanced**, and **Dense** change the number of automatically generated points.

- **Mean**, **Median**, and **Min volatility** determine the representative power used for an automatically generated point.
- **Show spline handles** displays the slope handles for a Smooth spline. **Move spline points in time** allows a point to be dragged horizontally as well as vertically.

Dragging a point corrects its position. A secondary-click on the graph adds a point **At Data** or **At Baseline**. A secondary-click on an existing point exposes **Lock** or **Unlock**, **Mark Linear** or **Unmark Linear**, and **Remove**. Marking neighboring points as linear makes the interval between them straight. Locked points are retained when automatic spline points are regenerated.

Balanced provides the middle spline-point density. Greater density increases local flexibility but can also follow noise or absorb part of an injection response.

Polynomial

Polynomial fits one polynomial across the complete thermogram and is suited to smooth global drift. **Degree** controls flexibility, from 0 to 24; the default is 12.

A polynomial baseline is least constrained at the beginning and end of the run, where a high degree can produce large deviations. Increasing the degree adds flexibility, but the baseline may then follow noise or part of an injection response.

Segmented

Segmented fits local constant, linear, or quadratic baseline behavior between integration regions and blends the local estimates across the run. It is suited to locally changing drift for which one global polynomial is too rigid. **Degree** selects the local behavior.

Local flexibility can follow genuine drift, but it also makes the result more dependent on the integration boundaries and on the baseline immediately around each peak.

Exclude integration regions from the baseline

When **Discard integrated regions** is enabled, data inside the current integration regions are excluded when the baseline is recalculated. Moving a boundary can therefore change both the integrated area and the estimated baseline.

> **CAUTION:**
> Boundaries should follow the observed response and a consistent processing rule, not the heat expected from a model.

Convert to a spline

A Polynomial or Segmented baseline can be converted to a **Smooth** or **Linear** Spline when the automatic baseline is a useful starting point for graphical editing. Conversion changes the baseline representation and creates editable spline points.

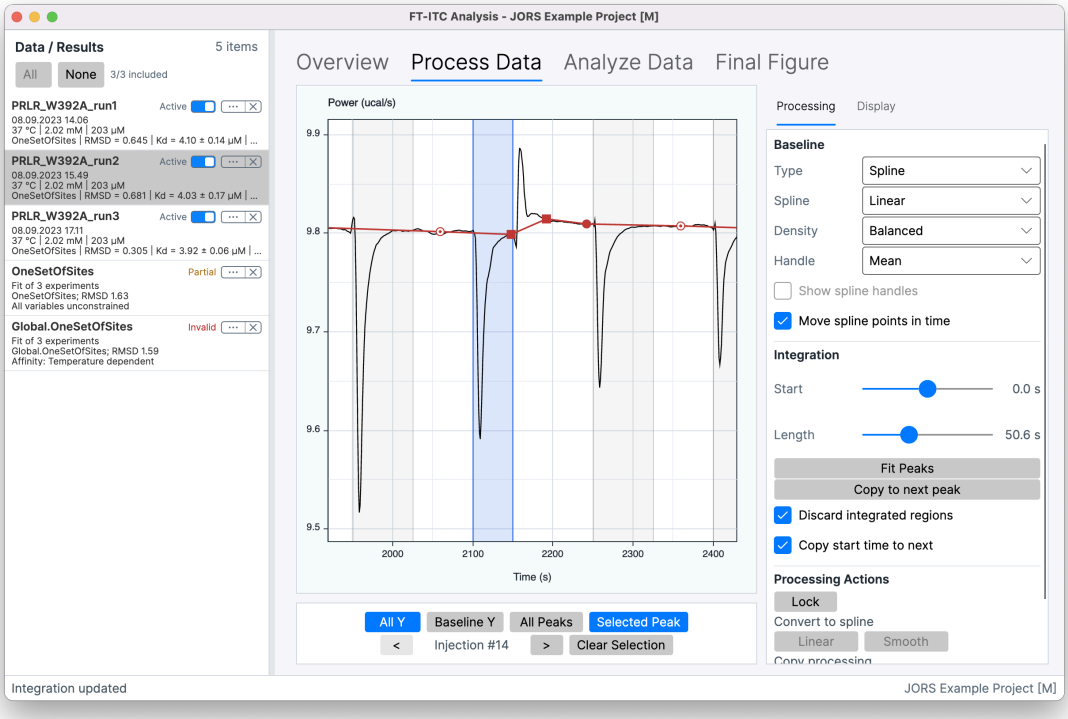
Conversion locks processing so that the converted points are kept. **Unlock** processing to edit the points. Moving, adding, removing, or individually unlocking a point retains the existing spline. Individual **Unlock** releases both the point position and slope locks; a Smooth spline recalculates

the released slope from the current points. Changing spline settings or integration regions, or running **Fit Peaks**, regenerates all spline points from the data and discards the converted points.

Conversion details

A Polynomial baseline is sampled at evenly spaced points; the point density setting controls how many. A Segmented baseline instead receives a point at each end of every fitted segment and one midpoint in each integration region where the blend between neighboring segments curves. A Linear conversion also adds a midpoint inside each quadratic segment. A Smooth conversion gives every point a fixed slope taken from the segments, so the fitted segments are kept exactly and only the curve across the integration regions is approximated. The point density setting does not apply to Segmented conversion.

Integration regions



Selecting an injection exposes its individual boundaries and navigation controls.

Each injection region has a start and an end boundary. **Start** sets the offset of the start boundary relative to the injection. Despite its label, **Length** sets the end boundary in seconds after the injection begins, rather than the duration measured from the selected start boundary. For example, **Start** = 2 s and **Length** = 20 s define an 18 s integration region.

Either boundary can be dragged in the graph or adjusted with the controls. The application constrains the start and end to a valid interval within the injection scope and preserves a minimum separation between them.

Estimate end points with Fit Peaks

Fit Peaks estimates the end of each injection response from the baseline-corrected thermogram. When **Discard integrated regions** is enabled, it recalculates the baseline using the updated regions.

When the endpoints settle, or a repeating set of endpoints is resolved, the application reintegrates the heats using the final regions and corresponding baseline. If peak fitting fails or does not find a stable result, the previous regions are restored. Peak shapes can vary across the titration, so review the boundaries, especially for the first injection.

The experiment records that its end points came from **Fit Peaks**, and the analysis report states this. Moving an end boundary, using the length control, or **Copy to next peak** marks the regions as set manually. Copying processing to other experiments runs **Fit Peaks** on each of them.

Copy a region to the next injection

Selecting an injection and choosing **Copy to next peak**, or pressing **Space**, copies its end boundary relative to the injection start and advances to the next injection. **Copy start time to next** includes the start boundary in that operation.

Propagate and lock processing

Active under **Copy processing** copies the selected experiment's processing to the other included experiments. This can replace existing processing when a destination is unlocked. **New** targets experiments without a baseline, whether included or not. In native macOS, **All** targets all other experiments, whether included or not.

Experiments without a raw thermogram are skipped; the status message reports how many.

Locked destinations are not overwritten. Copying from a locked source recalculates each destination's baseline and heats, then locks it. Copying from an unlocked source leaves destinations unlocked; choose **Lock** to protect them. Locking disables baseline, spline-point, integration-region, and peak-fitting edits; **Unlock** makes changes available again.

Integration calculation

CALCULATION:

$$q_{i,\mathrm{raw}} = \int [P(t) - b(t)]\,\mathrm{d}t$$

$$\Delta H_i = \frac{q_i}{n_i}$$

$$n_i = c_{\mathrm{syr}} V_i$$

The integral is evaluated between the injection's integration boundaries. Each recorded power value represents the trailing-period average ending at its timestamp, so fractional intervals at either selected boundary are included using the appropriate right-endpoint sample. $P(t)$ is differential power, $b(t)$ is the estimated baseline, and $q_{i,\mathrm{raw}}$ is the raw integrated heat. The heat q_i equals that raw value unless optional buffer subtraction changes it. c_{syr} is the syringe concentration, V_i is the injection volume, n_i is the injected amount, and ΔH_i is the molar heat.

Injection uncertainty

Each injection error bar shows an estimated ±1 standard deviation for that injection's molar heat: roughly, how much the calculated heat might vary because of local noise in the thermogram. It is calculated independently for every injection, so the bars can vary across a titration. It does not cover every source of experimental error. These processing dependent estimates can be used for weighted fitting.

Weight by injection error can use these estimates in the fitting objective. Concentration uncertainty is handled separately in supported resampling calculations. Both behaviors are described under Fitting calculation.

Noise calculation and limits

The calculation uses baseline-corrected samples around the injection. It combines an estimate of local power noise with the temporal correlation between neighboring samples, the integration-region length, and uncertainty in the baseline level. A longer or noisier region will therefore often have a larger estimated uncertainty. When Buffer Subtraction is applied, the independent target and reference heat uncertainties are combined.

Let $\sigma_{p,i}$ be local RMS power noise and T_{int} the integration duration in seconds. The heat SD combines integration-noise and baseline-level contributions:

CALCULATION:

$$\sigma_{q,i} = \sqrt{\sigma_{q,i,\mathrm{int}}^2 + \sigma_{q,i,\mathrm{bl}}^2}$$

$$\sigma_{q,i,\mathrm{int}} = \sigma_{p,i} \cdot \sqrt{\sum_{j,k} w_j w_k r^{|j-k|}}$$

$$\sigma_{q,i,\mathrm{bl}} = \sigma_{p,i} \cdot \frac{T_{\mathrm{int}}}{\sqrt{N_{\mathrm{bl,eff}}}}$$

w_j is the time each sample contributes to the heat, in seconds; r is the correlation between neighboring samples. $N_{\mathrm{bl,eff}}$ is the effective baseline-sample count accounting for correlated noise. Local noise is estimated from samples before integration and near the end of the injection interval; the latter can overlap integration. These are processing-derived heat SDs, not model-estimated parameter uncertainties.

For a molar-heat plot, the heat and its SD are both divided by the injected amount n_i defined above.

The correlation estimate uses only consecutive baseline sample pairs separated by at most twice the sampling interval. Both its product sum and its normalization use those same pairs, excluding a pair across a longer integration gap. Propagation assumes an AR(1) noise model, in which correlation decays geometrically with sample separation; this is a chosen approximation to the local noise.

NOTE:

If an injection has no processing-derived uncertainty estimate, its stored error is zero and no error bar appears. Integrated-heat imports have no thermogram from which to calculate this estimate. A zero or absent error bar does not establish that the injection has no uncertainty.

The bars do not include uncertainty in cell or syringe concentration, fitted parameters, or model-derived confidence bands. They also do not quantify baseline-model choice, integration-boundary choice, calibration error, other instrumental effects, or other systematic uncertainty. Consequently, they do not validate the selected processing or constitute confidence intervals for the true heat.

CHAPTER 06

Single-experiment fitting

Fit one experiment, configure model and uncertainty options, control injection inclusion, and interpret fit diagnostics.

Last verified: 2026-09-23

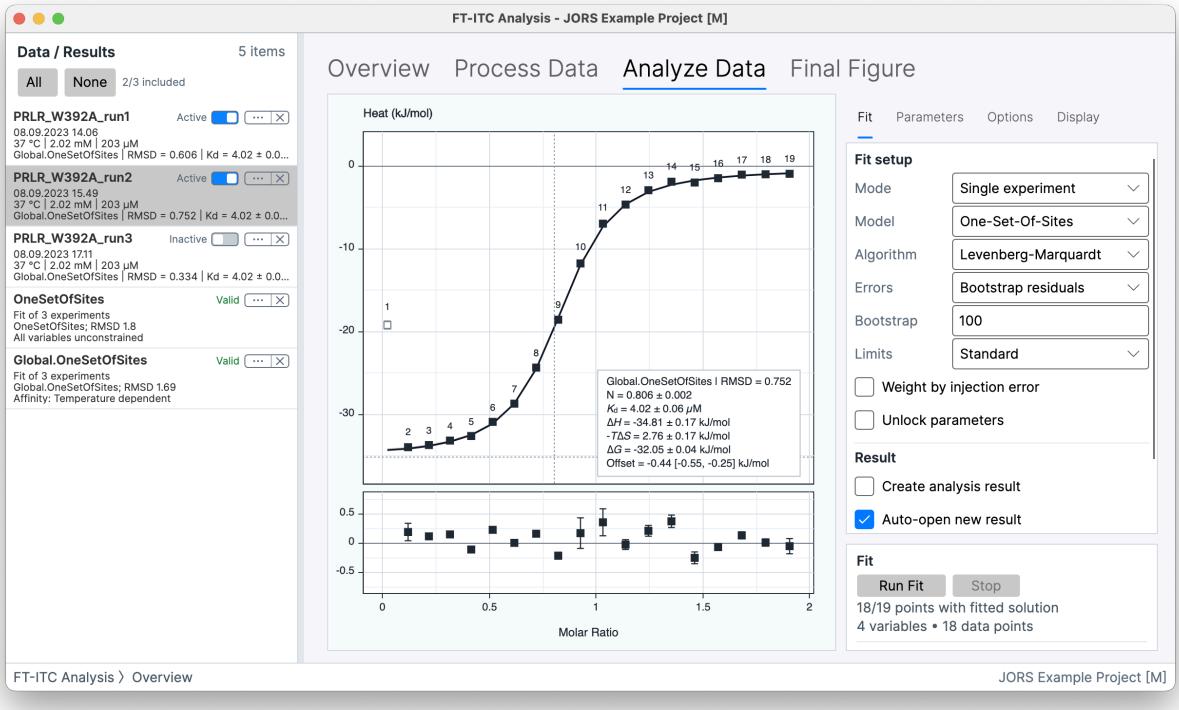
In **Analyze Data**, choose **Single experiment**, select a model, review starting values and options, select an uncertainty method, and choose **Run Fit**. Only included injections enter the fit. The concentrations, volumes, temperature, and model options define its scientific inputs.

Review residuals and parameter plausibility afterward. Successful optimizer termination alone does not establish that the model explains the chemistry.

The inspector has four tabs:

- **Fit** selects the model, optimizer, error-estimation method, limits, weighting, and result output. Its **Fit summary** always shows the selected experiment's fitted model, scope, RMSD, and fitted and derived parameters.
- **Parameters** shows the model parameters and their starting or fixed values.
- **Options** contains settings specific to the selected model.
- **Display** controls the fitted curve and diagnostic information shown in the graph. **Guides** draws lines at the fitted stoichiometry and enthalpy. Under **Parameters**, **Show overlay** adds a parameter box to the graph; **Derived parameters** and **Larger text** change only that box, not the fit summary.

In the native macOS Analysis view, the **Parameters** button above the graph shows or hides the parameter box, and **Larger Parameter Text** is in the **Analysis** menu on the right side of the toolbar.

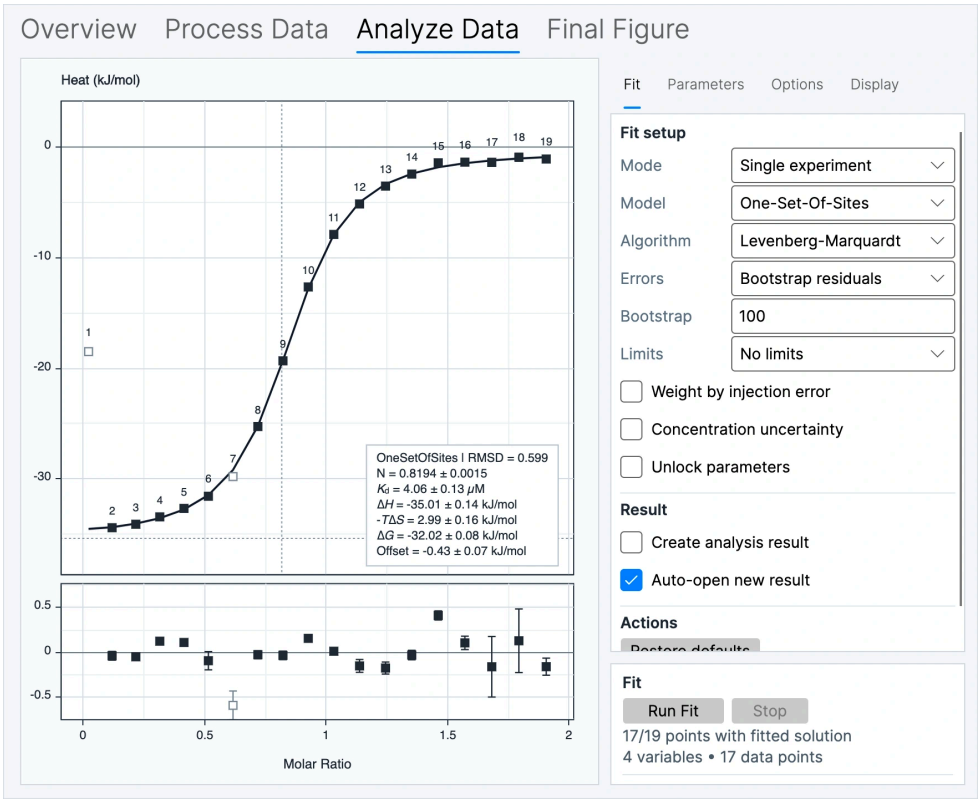


Analyze Data combines the fitted-heats graph and residuals with the controls for configuring and running the fit.

Multiple-experiment fitting uses additional experiment selection and parameter constraints; see Multiple-experiment fitting.

Injection inclusion

Selecting an injection point in the integrated-heats graph changes whether it is included in the fit. Excluded injections do not contribute to the objective function.



Excluded injections remain visible when Excluded points is enabled and can be selected for inclusion again.

Changing injection inclusion does not rerun the fit. The fitted curve and parameters continue to represent the previous fit until **Run Fit** is used again. An inclusion change can also invalidate an Analysis Result that contains the experiment.

Injection bookkeeping: MicroCal, Ideal continuous mixing, and Discrete displacement

Injection bookkeeping determines the concentrations used by the fit and accounts for reaction heat carried out of the active cell by displaced solution. It does not change measured peak areas or baseline integration.

- **Discrete displacement** is the default starting point for ordinary pulse injections. It assumes that solution is displaced before appreciable mixing, then the remaining cell contents mix with the injected solution.
- **Ideal continuous mixing** assumes mixing throughout the injection. Consider it for injections slow relative to cell mixing or to compare the opposite mixing limit.
- **MicroCal** uses the documented MicroCal concentration and displaced-heat approximations. Choose it when reproducing or comparing MicroCal analyses.

No method is assumed to be empirically superior. Injection bookkeeping is separate from dilution-heat background models and between-reload back-mixing.

Preferences > Processing > Injection bookkeeping selects the default for new data. **Experiment Details > Injection bookkeeping** explicitly switches an ordinary experiment and invalidates its fits without reintegrating measured heats. Changing the preference or editing a name/comment does not switch existing data. Older projects retain their saved concentrations and historical heat behavior; if their method is unknown, the selector displays **Saved processing — unchanged**. Choose a method explicitly before recalculating unknown saved concentrations. Rebuild tandem experiments through the tandem tool to change their method.

Concentrations and injection heat

For MicroCal, let u = cumulative injected volume / active cell volume, M_0 be the initial cell concentration, and C_s the syringe concentration. Starting with no ligand in the cell:

CALCULATION:

$$M = M_0 \frac{1 - u/2}{1 + u/2}$$
$$X = C_s u(1 - u/2)$$

The first line estimates how much macromolecule remains; the second estimates how much injected ligand is in the cell.

These are approximate concentration curves for the **MicroCal** bookkeeping method. They describe calculated concentrations, not measured curves.

Every delivered injection contributes to cumulative volume, including injections excluded from fitting. The ligand equation is the approximate expression used by FT-ITC for MicroCal processing. The displaced-volume assumptions remain approximate. For a tandem experiment, this calculation stops when cumulative injected volume reaches twice the active cell volume; ideal continuous mixing has no corresponding cumulative-volume limit. Existing projects retain saved concentrations and fits until concentration reprocessing.

For one injection, let v be injection volume, V active cell volume, and Q the equilibrium binding heat content of the cell in joules. The predicted injection heat q depends on the bookkeeping method.

MicroCal:

CALCULATION:

$$q = Q_{end} - Q_{start} + \frac{v}{V}\frac{Q_{start} + Q_{end}}{2}$$

The heat change is adjusted for displaced solution using the average of the heat content before and after the injection.

Ideal continuous mixing:

CALCULATION:

$$q = Q_{\text{end}} - Q_{\text{start}} + \frac{v}{V}\frac{Q_{\text{start}} + 4Q_{\text{mid}} + Q_{\text{end}}}{6}$$

This averages the displaced heat along the mixing path, including the state halfway through the injection.

Discrete displacement:

CALCULATION:

$$q = Q_{\text{end}} - (1 - \frac{v}{V})Q_{\text{start}}$$

Only the fraction of the original cell mixture left after displacement contributes its starting heat.

For Discrete displacement, concentrations advance after each injection as follows:

CALCULATION:

$$M_{\text{end}} = (1 - \frac{v}{V})M_{\text{start}}$$
$$X_{\text{end}} = (1 - \frac{v}{V})X_{\text{start}} + \frac{v}{V}C_{\text{s}}$$

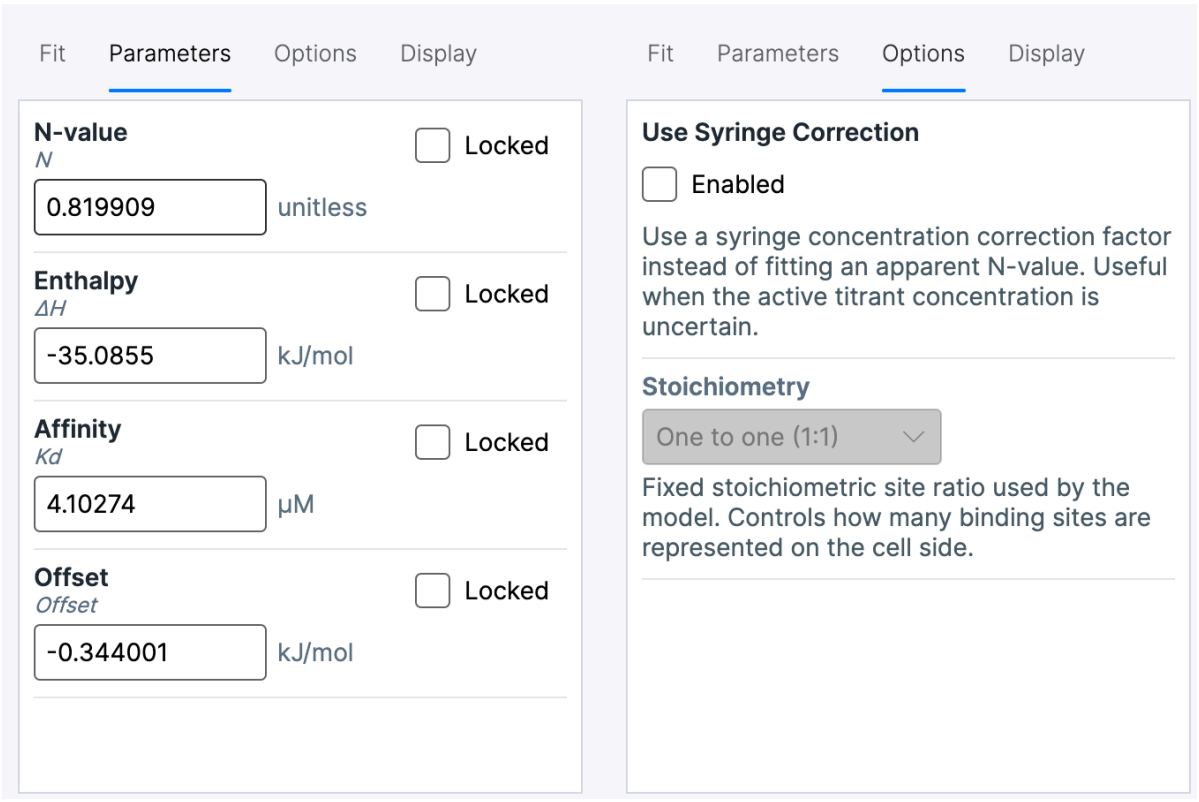
The same fraction of each starting concentration remains, while the injected solution adds ligand from the syringe.

Discrete displacement follows the formalism described by Freire, Schön and Velazquez-Campoy (2009). Across a fixed-syringe segment, the retained fractions are multiplied across the injections. Each injection must be smaller than the cell volume, although the total volume injected over the experiment can exceed it. The heat correction uses the cell's pre-injection heat content, which differs from the MicroCal convention and can change fitted values.

For ideal continuous mixing, the midpoint is evaluated halfway through each injection on the changing concentration trajectory. The MicroCal option retains its own heat convention.

The ideal continuous mixing heat calculation is inspired by Dumas (2022), but uses FT-ITC's finite-injection approximation rather than that paper's imperfect-mixing model. It does not model kinetic single-injection experiments. The calculation uses a fixed integration rule without adaptive refinement, so unusually large injections or sharp transitions need additional scrutiny.

Parameters and model options



Parameters exposes parameter values and locks; Options contains settings specific to the selected model.

The application generates initial parameter values from the experiment and can reuse an attached fitted solution where applicable. Entering a value replaces the generated starting value for that parameter. Values are displayed in the current application units. Clearing a value returns the parameter to automatic initialization.

Locked holds a parameter at its displayed value during the primary fit. An unlocked parameter is adjusted by the optimizer. Locked values remain part of the model and affect every other fitted parameter even though they are not estimated by that fit.

Analysis choices are retained separately for the available fitting modes and models, except that changing models resets global constraints to **Independent**. **Restore defaults** clears the stored analysis inputs and reloads the fitting controls from the current preferences. It also resets the inspector's **Unlock parameters** setting. The action does not change preferences, including saved limits, result-output options, and display settings.

Saving a project or applying Preferences preserves the solver and uncertainty choices in the Analysis inspector. Their saved defaults take effect at application startup or through **Restore defaults** in Analysis.

The **Limits** control selects a common parameter-bound policy:

- **Standard** uses the normal parameter bounds.
- **Expanded** permits a wider parameter range.

- **No limits** widens the configured bounds substantially. For affinity, **Expanded** allows K values 20 times lower to 20 times higher than Standard, and **No limits** allows K values 2000 times lower to 2000 times higher. These ranges are widened in $\log_{10} K$ space.

Negative $\log_{10} K$ values represent positive association constants below 1 in the application's affinity units. In multiple-experiment fits, globally constrained affinities controlled by fitted parameters are checked to ensure both K and K_{d} remain positive and finite. A candidate outside that numerical range is rejected; valid converted values can lie beyond the selected local affinity limits.

A fitted value at a bound is not an interior estimate. It indicates that the reported value depends on the selected bound as well as on the data and model.

Initial guesses outside the selected limits are adjusted to the nearest limit when you choose **Run Fit**. Locked values are preserved, and globally determined values continue to follow their shared constraints.

Fit execution and diagnostics

Run Fit starts the analysis. **Stop** requests cancellation of the active calculation.

While the analysis runs, the status bar identifies the current calculation: fitting, binding assessment, or uncertainty estimation. For independently fitted experiments, it shows the experiment name and position. The secondary status counts completed refits or profile searches.

When the analysis ends, the status bar shows the outcome and total time, followed by the uncertainty outcome. Check a fit that stopped at a limit, or uncertainty reported as incomplete, unavailable, or skipped, before using the result.

The **Display** tab exposes complementary diagnostics:

- **Fit line** shows the curve calculated from the current fitted solution.
- **Residuals** show the difference between each included observation and the fitted curve.
- **Error bars** show the processing-derived integration uncertainty.
- **Confidence band** shows uncertainty around the fitted curve when the solution contains suitable resampling results.
- **Excluded points** shows injections that do not contribute to the current fit.

These displays describe different aspects of the fitted solution. A small RMSD does not rule out systematic residual structure, poorly identified parameters, or dependence on model assumptions.

Store the fitted solution

With **Create analysis result** disabled, a successful single-experiment solution remains attached to its Experiment Data and is available in that experiment's analysis and figure workflows.

With **Create analysis result** enabled, a usable completed fit also creates a separate Analysis Result containing the experiment, model, fit settings, and solution. **Auto-open new result** determines whether the new result workspace opens immediately. A stopped or unusable fit does not create or replace an Analysis Result.

Fit availability and non-convergence

A single-experiment analysis requires processed or imported heats and at least three included injections with usable numerical values. The binding models also require nonzero cell and syringe concentrations. **Dissociation** uses the syringe concentration and does not require a macromolecule concentration in the initially buffered cell.

An unavailable model, failed fit, solution at a parameter bound, or large number of failed refits can reflect missing or degenerate heat and concentration information, the supplied starting values, the selected limit policy, the optimizer's interaction with the fitting surface, model complexity, or weak parameter identifiability. Resampling can fail even when the primary fit succeeds because each refit presents a different or reduced dataset to the same model. ## Models

Choose a model based on what is in the cell and syringe and what interactions are plausible. Adding parameters can improve a curve's fit even when the extra binding process is unsupported, so compare residuals and parameter uncertainty as well as the curve. The descriptions below state what each model assumes and reports.

INTERPRETATION:

When comparing fitted results from different programs, check the binding model, concentration and stoichiometry definitions, injection bookkeeping, and background-heat or offset treatment. Using the same injection convention does not necessarily make fitted parameters directly interchangeable.

One-Set-Of-Sites

One-Set-Of-Sites represents one class of equivalent, independent binding sites. It fits stoichiometry, dissociation constant, binding enthalpy, and an injection-heat offset. The solution also reports thermodynamic quantities derived from the fitted affinity and enthalpy.

The **Options** tab provides **Use Syringe Correction** and **Stoichiometry**. Without syringe correction, the fitted **N-value** represents the apparent site stoichiometry. With syringe correction enabled, **Stoichiometry** fixes the number of cell-side sites and **Correction factor** (α) scales the delivered titrant concentration. Enter or lock α in the parameter row; set the fixed site ratio under **Stoichiometry**.

The model cannot by itself distinguish concentration uncertainty from other effects that change an apparent stoichiometry.

Calculation and assumptions

Let M be the total macromolecule concentration, N the number of sites per macromolecule, X the total ligand concentration, x the free ligand concentration, and K_d the dissociation constant. A smaller K_d means stronger binding. The model finds x by balancing free and bound ligand, then calculates the heat content Q of the cell:

CALCULATION:

$$f = \frac{x}{K_\mathrm{d} + x}$$
$$X = x + NMf$$
$$Q = VNMf\,\Delta H$$

The fraction f tells us how many sites are occupied. The middle line counts free ligand plus ligand on the sites. The last line multiplies the amount bound in cell volume V by the heat per mole bound, ΔH. Injection heat then follows the selected bookkeeping calculation above.

Two-Sets-Of-Sites

Two-Sets-Of-Sites represents two independent classes of sites. It fits separate stoichiometries, dissociation constants, and enthalpies for the two classes, together with a shared injection-heat offset.

Shared N-Values makes the two site classes use the same fitted stoichiometry. **Use Syringe Correction** instead fixes the first and second **Stoichiometry** values and fits one active syringe-concentration factor, `alpha`.

With syringe correction on, set the two fixed stoichiometries independently. **Shared N-Values** applies only when syringe correction is off.

The two site labels are interchangeable: exchanging all parameters assigned to site 1 and site 2 describes the same physical model. A lower RMSD or optimizer objective alone does not establish that two distinguishable binding processes are supported by the experiment.

Calculation and assumptions

Each class has its own site count N, affinity K_d, and binding enthalpy ΔH. With free ligand concentration x, the two classes share the same supply of ligand:

CALCULATION:

$$f_j = \frac{x}{K_{\mathrm{d},j} + x}, \quad \text{for } j = 1 \text{ or } 2$$
$$X = x + M(N_1 f_1 + N_2 f_2)$$
$$Q = VM(N_1 f_1 \Delta H_1 + N_2 f_2 \Delta H_2)$$

Each f is the occupied fraction for one site class. The ligand balance finds one free concentration that satisfies both classes. Their heat contributions add to give the cell heat content; injection bookkeeping converts its change to a predicted heat.

Sequential Binding Sites

Sequential Binding Sites represents two, three, or four ordered binding steps on a macromolecule in the cell. **Sequential binding steps** in the **Options** tab selects the number of steps. The model fits one macroscopic stepwise association constant and one molar step enthalpy for each transition, together with the ordinary molar injection-heat offset. It does not fit an N-value or syringe activity.

The reported K_i values are phenomenological, macroscopic step constants for the ordered transitions $M \rightarrow MX \rightarrow MX_2$ and so on. They are not microscopic intrinsic site constants. Step numbers therefore have a physical order, so fitted steps are never sorted or treated as interchangeable.

The macromolecule must be in the cell and ligand in the syringe. Reverse titrations with macromolecule in the syringe are outside this model. Multi-step fits can be weakly identifiable, especially when the concentration window does not populate every transition. Inspect parameter bounds, residuals, bootstrap uncertainty, and parameter correlations; an improved RMSD does not by itself establish the selected number of sequential steps.

Sequential-state calculation

For step count n, let $\beta_0 = 1$, $\beta_i = \prod_{j=1}^{i} K_j$, and let x be the free ligand concentration. The state weights and fractions are

CALCULATION:

$$w_i = \beta_i x^i$$

$$F_i = \frac{w_i}{\sum_{j=0}^{n} w_j}$$

$$\bar{\nu} = \sum_{i=0}^{n} i F_i$$

$$X_t = x + M_t \bar{\nu}$$

Each state receives a weight from its binding steps and the available free ligand. Dividing by the total weight gives the fraction in that state. The average number of bound ligands, $\bar{\nu}$, then balances free and bound ligand against the total.

Here M_t and X_t are total macromolecule and ligand concentrations in the cell. The model solves the ligand balance as part of the fit and calculates the cell heat content from the population of every sequential state:

CALCULATION:

$$Q = VM_{\mathrm{t}} \sum_{i=1}^{n} F_i \left(\sum_{j=1}^{i} \Delta H_j \right)$$

For each state, add the heats of all steps needed to reach it, multiply by the fraction in that state, and sum across states. Multiplying by the macromolecule amount in the cell gives its heat content.

Competitive Binding

Competitive Binding represents titration of a target ligand into a macromolecule that is initially in equilibrium with a prebound ligand in the cell. It fits the target ligand's stoichiometry, dissociation constant, binding enthalpy, and injection-heat offset.

The **Options** tab requires the pre-equilibrated competitor's **Total competitor** concentration, **Ligand Affinity**, and **Ligand Enthalpy**. **Total competitor** includes both free and bound competitor in the cell. Each option can use an experiment attribute instead of a manually entered value. For affinity and enthalpy, add a **Competitor properties** attribute and select a one-set-of-sites Analysis Result to supply K_{d}, ΔH, and their uncertainty. Different experiments in a global fit can select different source results: eg if using a different competitor. The model also provides **Use Syringe Correction** and **Stoichiometry** with the same concentration-factor interpretation as One-Set-Of-Sites.

With **From attributes** selected, the next fit uses each experiment's attribute value; the manually entered option is used when **From attributes** is off. Every included experiment needs the required attributes, or the fit cannot start. The Analysis Result **Model** page identifies attribute-supplied options and the values used for each experiment. **Update Result** reads current attributes; an existing result does not update when attributes change and retains the values used in its earlier fit until updated.

The **Total competitor** attribute also accepts an SD for its concentration.

If the source result has temperature dependence, its values are evaluated at the experiment temperature. Otherwise its summary temperature is used, with a notice when the temperatures differ. Removing a source after fitting does not erase values already used by a result; a source removed before its values were used must be restored or replaced. Updating the source result marks dependent results out of date until they are fitted again. Residual bootstrap samples the supplied affinity and enthalpy uncertainties separately, as it does for manually entered model options.

The **Ligand Affinity** and **Ligand Enthalpy** labels describe the pre-equilibrated competitor's properties. The fitted target affinity and enthalpy depend on those supplied properties. The competitor properties are model inputs, not quantities independently determined by this fit.

The reported apparent target K_{d} includes a competition factor calculated from the initial free competitor concentration. This calculation accounts for competitor already bound to cell sites rather than substituting the total competitor concentration. The apparent target K_{d} can therefore differ from its intrinsic value.

Calculation and assumptions

The target (A) and competitor (B) compete for the same sites. Let a and b be their free concentrations, $K_{a,A}$ and $K_{a,B}$ their association constants, and S the total site concentration. The model solves for the two free concentrations so that the available ligand is accounted for:

CALCULATION:

$$D = 1 + K_{a,A}\,a + K_{a,B}\,b$$
$$f_A = \frac{K_{a,A}\,a}{D}$$
$$f_B = \frac{K_{a,B}\,b}{D}$$
$$A_{\text{total}} = a + S f_A; \quad B_{\text{total}} = b + S f_B$$
$$Q = VS(f_A \Delta H_A + f_B \Delta H_B)$$

D counts the empty and occupied possibilities for a site; f_A and f_B are the occupied fractions. The two ligand balances count free plus bound molecules. Each bound population contributes its own heat. The competitor's affinity and enthalpy are supplied inputs; the target's are fitted.

Dissociation

Dissociation describes dimers in the syringe solution separating into free components when injected into a cell initially containing buffer. Dilution shifts the equilibrium toward free components. This includes a 1:1 complex of two different components (a heterodimer) when they are present in equal amounts. The model fits an effective dissociation constant, the heat per mole of dimers formed, and an injection-heat offset.

The integrated-heats and residual plots use the total titrant concentration in the cell after each injection, in µM. This concentration is calculated from the entered concentrations and injection bookkeeping.

For an equimolar complex $A + B \rightleftharpoons AB$, enter the syringe concentration as the **sum of the concentrations of A and B units** (twice the concentration of an equimolar AB preparation). Then $F = [A] + [B]$ and $P = [AB]$. Because $[A] = [B] = F/2$, the usual heterodimer association constant $K_{a,AB} = [AB]/([A][B])$ is **four times** the model's $K_{a,\text{model}}$; the usual heterodimer K_d is one quarter of the fitted K_d. The fitted ΔH_{assoc} is per mole of AB formed. The model does not represent unequal component amounts, separate component concentrations, or other oligomerization schemes. It has no stoichiometry or syringe-correction options.

Dissociation calculation

CALCULATION:

$$K_{\mathrm{a,\,model}} = \frac{P}{F^2}$$
$$C = F + 2P$$
$$Q = VP\,\Delta H_{\mathrm{assoc}}$$

F is the concentration of free components counted together, P is the concentration of dimers, and C counts two components per dimer. The equilibrium determines how much remains dimerized at each dilution. The cell heat content is the amount of dimer times the heat of forming it. Dilution drives dimers apart, so dissociation heat has the opposite sign from the fitted association enthalpy. Injection bookkeeping also accounts for associated species carried in from the syringe.

With **MicroCal** bookkeeping, Dissociation uses MicroCal's own dimer-dissociation equation instead of the averaged displaced-heat formula used for the binding models. The displaced solution leaves with the cell content from before the injection, while concentrations still follow the MicroCal equations. With v the injection volume, V the active cell volume, and P_{syr} the dimer concentration in the syringe:

CALCULATION:

$$q = \Delta H_{\mathrm{assoc}}\,[VP_{\mathrm{end}} - (V - v)P_{\mathrm{start}} - vP_{\mathrm{syr}}]$$

The last term counts the dimers delivered by the injection as part of the starting amount, so heat comes only from the change in dimer amount, mainly dissociation of injected dimers as they are diluted. MicroCal's concentration and heat equations treat displaced solution differently, so they do not balance exactly. Use this combination to compare with MicroCal results.

Thermodynamic relationships

The application derives thermodynamic quantities from the fitted affinity and enthalpy. Temperature T is expressed in kelvin.

CALCULATION:

$$K_d = \frac{1}{K_a}$$
$$\Delta G = RT\ln(K_d) = -RT\ln(K_a)$$
$$-T\Delta S = \Delta G - \Delta H$$

Here, $R = 8.31446261815324\ \mathrm{J\,mol^{-1}\,K^{-1}}$ is the molar gas constant. Its exact SI value is $R = k_B N_A$, where $k_B = 1.380649 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ is the Boltzmann constant and $N_A = 6.02214076 \times 10^{23}\ \mathrm{mol^{-1}}$ is the Avogadro constant; both are exact SI defining constants (NIST CODATA values). The application represents this value at double precision.

The final relationship matches the **–TΔS** quantity reported by the application.

Fitting calculation

Algorithm

Levenberg-Marquardt uses local derivative information and can be efficient when the starting values describe a suitable region of the fitting surface.

Nelder-Mead is a derivative-free simplex optimizer. It provides an alternative when the derivative-based method has difficulty converging from the chosen starting values.

Weight by injection error

Weight by injection error uses the integration uncertainty estimated during thermogram processing when calculating the fitting objective. An injection with a larger estimated error has less influence on the fitted parameters than one with a smaller estimated error. Choose this only when those processing-derived error estimates are suitable for the dataset.

RMSD summarizes the typical size of the heat differences between the observed points and the fitted curve. It is always calculated from the unweighted residuals, including after a weighted fit. The result tooltip reports the optimizer objective separately: the raw residual sum of squares for unweighted fitting or the standardized residual sum of squares for weighted fitting. Weighting can change the fit even though the displayed RMSD remains an unweighted summary.

For a multiple-experiment result, the displayed global RMSD is pooled across every included injection in every member experiment. Each member row retains its own local RMSD, so the global value remains comparable when members contain different numbers of included injections.

Weight by injection error is available only when every included injection has a finite, positive peak-area SD. The application checks this again before fitting. An injection without an estimated SD can be excluded from the fit, or its thermogram can be reprocessed to obtain an estimate.

The weighting describes the application's processing-derived uncertainty model. It does not account for every systematic source of experimental or processing uncertainty.

Objective calculation

CALCULATION:

$$r_i = q_{i,\text{obs}} - q_{i,\text{model}}$$

$$\text{RMSD} = \frac{\sqrt{\sum r_i^2}}{N}$$

$$\text{weighted objective} = \sum \left(\frac{r_i}{\sigma_i}\right)^2$$

Only included injections enter these sums, and N is their number. The value σ_i is the processing-derived uncertainty for injection i; weighting changes the fitting objective but does not remove systematic uncertainty.

Parameter uncertainty

The reported value is the best fit to the original data. Choose **Errors** to estimate uncertainty or sensitivity afterward:

- **None:** keep the primary fit without parameter uncertainty.
- **Bootstrap residuals:** estimate variation under resampled residual noise.
- **Leave-one-out:** assess sensitivity to omitting an injection or, for a global fit, an experiment.
- **Profile likelihood:** search for a 95% interval while refitting the remaining free parameters.

Bootstrap sets the requested number of residual-bootstrap iterations. It is enabled and used only for residual bootstrap; leave-one-out and profile likelihood have deterministic schedules and do not use this count. Only included injections supply residuals, and only retained usable refits enter the parameter distributions. Because residual-bootstrap sampling is with replacement, one residual can occur more than once in a synthetic dataset while another may not occur at all. The fit status distinguishes successful and failed refits.

Correlation diagnostics distinguish all attempted refits, those with usable optimizer results, and complete refits with finite values for every displayed parameter. If many refits fail, the retained set may not represent the full range of resampled outcomes. Increasing the requested count improves Monte Carlo precision but does not resolve systematic refit failures.

Rerunning a bootstrap can give slightly different uncertainty estimates because it draws a new random sample.

For a stored bootstrap result, **Update Result** offers the stored request or another supported count. It creates a fresh fit and bootstrap rather than appending samples. Canceling or retaining no usable refits preserves the previous result.

Residual-bootstrap calculation

Residual bootstrap samples from the included injections' primary-fit residuals and adds each draw to a best-fit prediction. An unweighted fit centers and samples raw residuals, assuming roughly equal residual spread across injections. A weighted fit divides by each injection's peak-area SD before centering and sampling, then multiplies each draw by the target injection's SD.

Fitted residuals underestimate noise because fitting absorbs some variation. The centered residuals are therefore multiplied by $\sqrt{n/(n-p)}$, where n is the number of included injections and p the number of free fitted parameters; locked parameters are not counted. This matches the residual variance estimate $\mathrm{RSS}/(n-p)$, where RSS is the residual sum of squares. Without the correction, small-dataset intervals would be too narrow. Shared global fits use pooled counts; independent experiments use their own. In either case, n must exceed p.

The primary best-fit parameter remains the reported value. For a parameter with best-fit value $\hat{\theta}$ and values θ_b from B retained refits, the application summarizes the bootstrap distribution as follows:

CALCULATION:

$$\mathrm{SD} = \frac{\sqrt{\sum_b (\theta_b - \hat{\theta})^2}}{B}$$

$$95\%\ \mathrm{CI} = [P_{2.5}(\{\theta_b\}), P_{97.5}(\{\theta_b\})]$$

SD is the root-mean-square deviation of the retained refits from the primary best fit. The confidence limits are the 2.5th and 97.5th percentiles of the retained refit distribution itself, so they need not be equally spaced around the best fit. Neither calculation replaces the best-fit value with the bootstrap mean or median.

The uncertainty display can show SD, the 95% confidence interval, both, or select between them automatically. This presentation rule is described under Uncertainty and evaluation temperature.

Leave-one-out calculation

Leave-one-out performs one deterministic refit per included injection in a single-experiment analysis, or per omitted experiment in a global fit. Concentrations, uncertain model options, and parameter locks remain at their primary-fit values, so the spread measures sensitivity to omission under those fixed inputs.

Profile-likelihood uncertainty

Profile likelihood estimates parameter uncertainty by asking how far a fitted parameter can be moved away from its best-fit value before agreement with the data crosses the chosen threshold. Profile likelihood does not create a refit ensemble or a confidence band.

For each trial, the selected parameter is fixed while the other free parameters readjust. Searching both directions allows compensation between parameters. Independent fits use their local objective; shared global fits use the complete objective and its free coordinates.

The 95% confidence limits are the parameter values at which the best achievable fit reaches a statistically defined 95% threshold. FT-ITC uses an F-calibrated threshold based on the number of observations and fitted parameters. The confidence interval is therefore determined by the shape of the fit surface rather than by assuming that parameter uncertainty is symmetric or normally distributed.

A complete interval is reported only when the threshold is crossed on both sides of the best-fit value. If the search reaches a parameter limit first, that side is recorded as censored and has no confidence endpoint. For a complete interval, the displayed `value ± SD` uses a symmetric display equivalent calculated from the asymmetric interval; it is not a sample standard deviation. Save the result in an `.ftxtc` project to preserve the profile diagnostics.

After **Update Result**, check whether the completion message says the previous profile result was retained rather than replaced.

Concentration uncertainty

With **Concentration uncertainty** enabled in the **Fit** tab, the concentration SDs entered in **Details...** are propagated through residual-bootstrap calculations. The control is initialized from the corresponding preference and is active only for residual bootstrap; leave-one-out and profile likelihood keep primary concentrations fixed. Each nonzero fractional SD is the arithmetic standard deviation relative to the entered concentration.

Bootstrap concentration variations use a positive lognormal multiplier with mean 1 and the entered fractional SD c. Specifically, $\sigma_{\log}^2 = \ln(1 + c^2)$, $\mu_{\log} = -\sigma_{\log}^2/2$, and the multiplier is $\exp(\mu_{\log} + \sigma_{\log} Z)$ for a standard-normal Z. This chosen distribution keeps sampled concentrations positive while preserving the entered arithmetic mean and SD; it does not change the concentrations used for the primary best fit. If a concentration has no entered SD, that concentration remains fixed in bootstrap refits. These uncertainties affect the synthetic experiment concentrations used for bootstrap refits.

Displayed parameter uncertainty

Uncertainty is shown in the same units and parameter form as the reported value. For example, affinity is fitted as $\log_{10}(K_a)$ but normally displayed as K_d; its SD is propagated through that conversion, and its percentile limits are transformed and ordered on the display scale. The central value remains the best fit to the original data, and this nonlinear conversion can make the displayed interval asymmetric.

Standard and Strict precision round halfway values away from zero, so mirrored positive and negative estimates and interval limits round symmetrically.

Quantities calculated from more than one reported parameter, such as $-T\Delta S$, use the application's uncertainty-propagation rules for that calculation. Their displayed limits are therefore not necessarily the percentiles that would be obtained by recalculating the complete derived quantity independently for every bootstrap refit. The **Automatic** SD-or-CI decision is applied after transformation or propagation, separately for each displayed quantity.

Unlock parameters during error estimation

Locked parameters remain fixed during the primary fit. With **Unlock parameters** enabled, copies of those parameters are unlocked only for residual-bootstrap refits and can vary in the resampled solutions. Leave-one-out and profile likelihood always preserve the primary-fit locks.

This setting does not change or rerun the primary best fit. It changes only the parameter state used by the repeated error-estimation fits, and it has no effect when no fitted parameter is locked.

CHAPTER 07

Multiple-experiment fitting

Multi-dataset fitting across processed Active experiments, including constraints, shared options, diagnostics, and combined results.

Last verified: 2026-09-23

Use **Multiple experiments** when several experiments should be analysed with the same model. Mark at least two processed experiments **Active** in **Analyze Data**. You can let each experiment have its own parameter values, or tell the fit that a supported parameter has a shared or temperature-dependent relationship across the experiments. This is useful when the experiments test the same interaction under different conditions.

When every constraint is **Independent**, the members are fitted independently. When one or more supported constraints are active, the solver performs one global optimization across the experiments. Each experiment remains a member of the combined **Analysis Result**, with member-specific parameters where the constraint state is **Independent**.

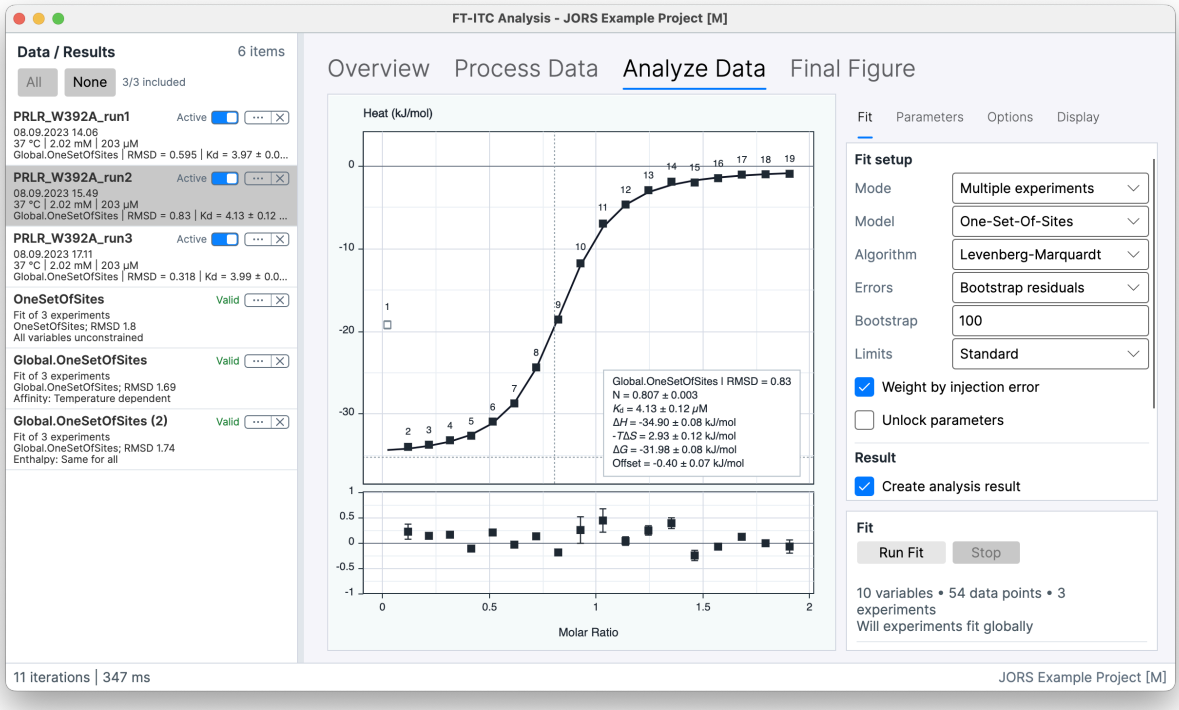
The model and model options apply across the active set. The resulting **Analysis Result** retains the member fits, any shared or temperature-dependent parameters, constraints, solver settings, diagnostics, and uncertainty output. Shared fitting controls are described in Single-experiment fitting.

Fit

In **Fit**, select the shared model, optimizer, uncertainty method, limits, and weighting, then choose **Run Fit**. Enable result creation to retain the collection. The controls work as described in Single-experiment fitting.

For a global fit, **Leave-one-out** checks the effect of omitting each experiment in turn. **Profile likelihood** evaluates the complete global fit, including its shared and member-specific free parameters. Independently fitted members use their own data and parameters for these calculations. Bootstrap count, concentration uncertainty, and parameter unlocking are available only with **Bootstrap residuals**.

The **Result** controls describe whether a completed fit is stored as an **Analysis Result** and whether that result opens automatically. **Run Fit** and **Stop** are the fit controls. The status area records termination state, RMSD, iteration count, elapsed time, and error-estimation outcome when applicable. A multiple-experiment analysis is ready only when every member has usable processed data and the set contains at least two members.



A shared enthalpy constraint connects the three Active experiments in one global fit.

Parameters

The **Parameters** tab contains the exposed fit parameters and, in multiple-experiment mode, the **Global constraints** section. Parameter rows show the current value, units, uncertainty information when available, and the parameter lock state. Values and locks control individual parameters. Global constraints define how those parameters are related across experiments.

The constraint states have these meanings:

STATE	MEANING
Independent	The parameter remains member-specific. Each experiment has its own fitted value.
Shared	One common value is fitted for every member in the set.
Temperature dependent	A supported parameter is represented across the temperature series by the relationship exposed by the model.
Thermodynamically linked	Affinity shares ΔG at the fixed fit reference and derives $\Delta G(T)$ from the selected enthalpy relationship, including ΔC_p when present.

The **Locked** state fixes an exposed parameter at its displayed value during fitting. **Locked** is not a constraint state and does not make a parameter common to the member experiments. A locked global value and a member-specific locked value therefore have different scopes.

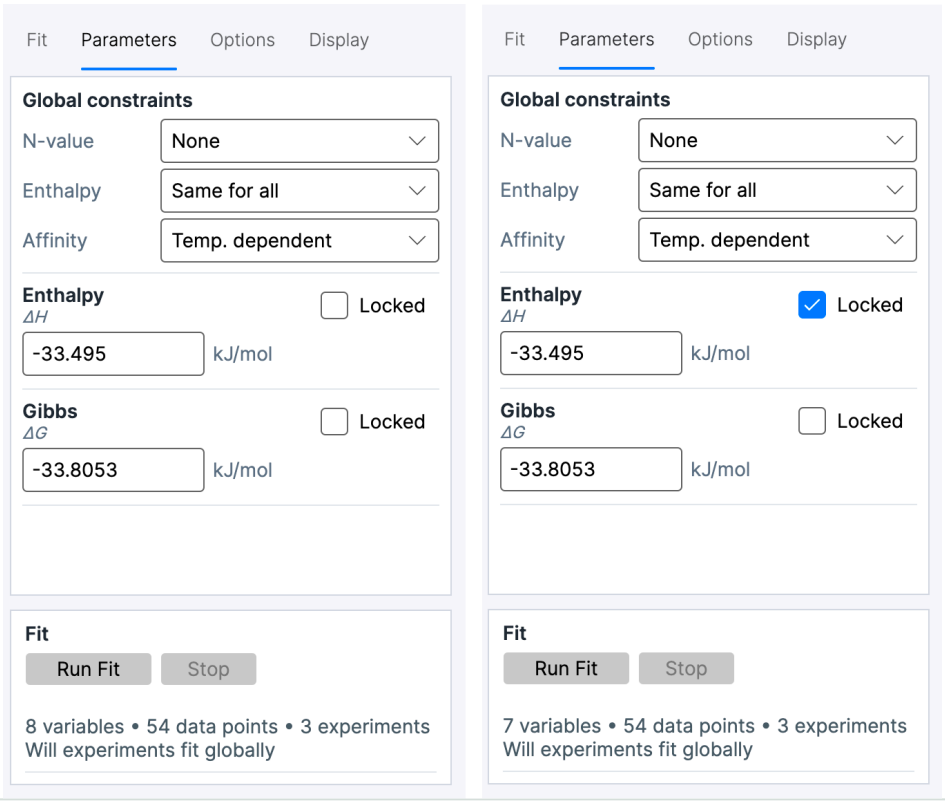
For the core binding parameters, the available relationship states are model- and data-dependent:

PARAMETER	AVAILABLE STATES
Affinity	**Independent**, **Shared Kd**, **Shared ΔG**, or **Thermodynamically linked**
Enthalpy	**Independent** or **Shared**; **Temperature dependent** is also available when the selected set exposes temperature dependence
N-value	**Independent** or **Shared**
Offset	**Independent** or **Shared**

The interface omits unsupported states for the current model. For affinity, **Independent** fits each member, **Shared Kd** fits one common dissociation constant, **Shared ΔG** shares one Gibbs-energy coordinate while K_d varies with temperature, and **Thermodynamically linked** derives the full temperature relationship from the selected enthalpy model. Profile linked intervals are approximate because fitted-coordinate correlations are omitted. Bootstrap and leave-one-out evaluate complete relationships, while the structuring analysis (Spolar–Record method) samples linked inputs independently in its Monte Carlo calculation.

For **Sequential Binding Sites**, the step count is one shared model option for the complete experiment set. The interface shows one **Affinity** constraint selector and one **Enthalpy** constraint selector, regardless of whether two, three, or four steps are active. The selected constraint applies to every active step while keeping a separate parameter for each step: **Same for all** shares K_1 across members, separately shares K_2, and so on; it does not force the different steps to equal one another. **Temperature dependent** likewise uses one ΔG_i per affinity step and one reference ΔH_i/$\Delta C_{p,i}$ pair per enthalpy step. Reducing the step count discards inactive step values and constraints; increasing it later creates new defaults for those steps.

Offset is an energy-per-mole-of-injectant correction. For each injection, its absolute heat contribution scales with the injected amount—the injection volume multiplied by the syringe concentration. With **Offset: Independent**, each experiment has its own fitted molar offset. With **Offset: Shared**, the **Parameters** tab exposes one common molar offset for the complete set. That shared value has its own **Locked** control; locking it fixes the common value for every member.



Linked temperature calculation

CALCULATION:

$$\Delta H(T) = \Delta H_{\mathrm{ref}} + \Delta C_{\mathrm{p}}\,(T - T_{\mathrm{ref}})$$

$$\Delta G(T) = (T/T_{\mathrm{ref}})\Delta G_{\mathrm{ref}} + (1 - T/T_{\mathrm{ref}})\Delta H_{\mathrm{ref}} + \Delta C_{\mathrm{p}}\,[(T - T_{\mathrm{ref}}) - T\ln(T/T_{\mathrm{ref}})]$$

$$K_{\mathrm{a}}(T) = \exp[-\Delta G/(RT)]$$

ΔH_{ref} is the enthalpy at the fixed fit reference, and ΔC_{p} is the equal-per-experiment regression slope when enthalpy is independent. The fit reference is fixed when the analysis is built; it is separate from the reporting temperature selected in preferences. A member's heat prediction continues to use its own fitted ΔH, while linked affinity uses the enthalpy trend. T and T_{ref} are absolute temperatures.

For independent enthalpies, the linked temperature relationship gives every experiment equal weight, including repeated temperatures and locked enthalpies. Linked Gibbs and entropy profile intervals propagate uncertainty through the enthalpy trend but do not account for correlations among fitted parameters; they are approximate. Bootstrap and leave-one-out recompute the relationship from their participating members. The reported best-fit values remain those from the original data.

Options

The **Options** tab contains the options exposed by the selected model. In multiple-experiment mode, a model option applies to every member. This includes model-specific concentration, stoichiometry, syringe-correction, or prebound-species settings when those options are exposed. The option values are preserved with the combined Analysis Result.

With **From attributes**, each member supplies its own attribute value under that common configuration. See Competitive Binding for competitor sources and the values captured by a fit.

An option that changes which parameters are exposed can also change the available parameter rows and constraint states. For example, model options that share or replace stoichiometry alter which N-values are independently represented. The available rows always describe the active model and option combination.

Display

The **Display** tab controls the graph presentation for the active member set. Its controls cover the fitted line, residuals, error bars, confidence band, point labels, parameter box, excluded points, automatic scaling, unified axes, fitted offset, and fit-line interpolation. The parameter box can separately show model, fitted, and derived parameters.

Display settings affect the graph presentation, not the model, the member data, or the fitted values. The graph can therefore show the independently fitted curves or global fit alongside each experiment's observations while the Parameters tab remains the source of the stored parameter and constraint state.

Combined Analysis Result

Open the combined **Analysis Result** to compare shared and member-specific values, then select members to inspect their fitted curves and residuals. The saved result retains the member set, model, constraints, solver and weighting settings, and uncertainty output.

The status and diagnostics distinguish a successful solver termination from the quality and uncertainty of the fit. RMSD, iteration count, weighting, error-estimation method, and the number of successful uncertainty refits describe how the combined result was obtained. Wide or asymmetric uncertainty, a member with a markedly different residual pattern, or a member solution with an invalid status remains visible as member-level information rather than being hidden by a combined value. Result interpretation and validity are covered in Results and advanced analyses.

CHAPTER 08

Results and advanced analyses

Analysis Result views, validity, uncertainty presentation, evaluation temperature, and conditional advanced analyses.

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An **Analysis Result** saves a fit so you can inspect and export it later. Start with the fitted values and graph, then check where the predicted heats differ from the measured heats. The uncertainty display describes how precisely the parameters were estimated under the chosen model; the validity indicator tells you whether the saved fit still matches the experiment data currently in the project.

For fit setup, see Single-experiment fitting and Multiple-experiment fitting. This chapter explains the saved result, its validity, and how to interpret its outputs.

Result views

The result view selector contains **Fit**, **Correlation**, and **Summary** for every Analysis Result. Thermodynamic temperature plots and parameter evaluation can show every active step in a sequential binding model. **Temperature** (temperature dependence and structuring), **Salt**, and **Protonation** appear only when the selected model defines those analyses and the member metadata satisfy their additional requirements.

Summary presents the combined parameter graph and result table. The table can show fitted values, derived values, and the selected uncertainty representation for each stored solution.

Molar-energy values use a common displayed unit; ΔC_{p} can use a different unit.

Select a member row to view its **Fit** and available **Correlation** information. Use **Preferences > General > Energy units** to choose joules or calories.

Advanced analyses can sample the stored values and uncertainty summaries from profile-likelihood results. This does not provide a bootstrap ensemble or a joint model of parameter covariance. Saved leave-one-out refits can provide an envelope on the integrated-heats graph.

Fit presents the saved fitted curve, residuals, error bars, confidence band, and excluded points for the selected member. The graph is read-only: it represents the stored solution and does not expose fit controls or alter the underlying experiment.

Correlation presents a matrix calculated from residual-bootstrap refits. Its availability, scope, and interpretation are described under Parameter correlation.

Result details

Open a saved result's details to edit its name and comments and inspect its saved fit information. On native macOS, **Result Details** opens as a sheet with three tabs:

- **Details** contains the name, read-only date, saved **Analysis operator**, comments, and a summary of the model, fit metrics, solver, uncertainty settings, and validity reasons.
- **Experiments** lists the result's member experiments, including their dates, temperatures, and solution status.
- **Actions** provides **Copy result table**, **Load solutions to experiments**, and **Select result experiments**. Selection includes the experiments belonging to this result and excludes the others in the current project. Copying uses the preferred energy-unit family and temperatures in Celsius.

Apply saves a nonblank name and the comments; **Cancel** discards those pending edits.

The saved **Analysis operator** records the operator associated with the fit when the result was created or last successfully updated. Results without a saved name show **Not recorded**.

Inspector tabs

The result inspector has four tabs with shared labels across the supported desktop versions: **Summary**, **Analysis**, **Experiments**, and **Model**.

The **Summary** tab contains the result identity, model, member count, RMSD, validity, information criteria, and solver diagnostics. Solver information includes algorithm, iterations, whether injection-error weighting was used, and the uncertainty method. Uncertainty status describes whether estimates are complete, incomplete, cancelled, unavailable, or intentionally skipped because no binding was detected.

Bootstraps shows the requested number of refits; its tooltip gives the retained count when it differs. Reports use complete retained replicates. Older results without a saved request show **Unknown**. For mixed independent results, eligible experiments retain their own uncertainty, but a complete combined set is unavailable. **Update Result** can rerun the stored request or use any offered count, including the same or a smaller count.

In the browser viewer, RMSD is shown as a saved unweighted display diagnostic in µJ, separate from the weighted fitting objective. When the saved convergence record contains it, **Molar RMSD (kJ/mol)** is shown separately: the result summary uses the saved global metric, while an individual fit uses that fit's saved metric. Missing or non-finite saved values remain unavailable; member values are not averaged to reconstruct the result metric.

Competitor properties in viewer metadata show captures saved on the experiment; these may differ from the values used by a particular saved fit.

The **Analysis** tab contains the result view selector, parameter evaluation, and the analysis-specific controls and outputs. It is also the location of the uncertainty display, correlation information, and evaluation-temperature presentation associated with the selected view. Energy units are chosen consistently from the displayed central values and are also used for graph axes, errors, fitted bands, tooltips, and parameter lists.

The **Experiments** tab lists the result members and their stored status and condition information, including member temperature. The row selected in the result table determines which experiment appears in **Fit**; this tab provides that experiment's details.

The **Model** tab shows the stored model options, locked parameters and their fixed values, and the active constraints. Constraints with state **Independent** are not listed as active global constraints. Affinity constraints are labelled **Independent**, **Shared Kd**, **Shared ΔG**, or **Thermodynamically linked** to describe the fitted relationship precisely; other parameters use **Shared** and, where supported, **Temperature dependent**.

Result validity

Validity answers whether the saved fit still describes the experiment inputs currently in the project. FT-ITC compares the result's saved inputs with the current member experiments, including their temperature, concentrations, cell volume, injection bookkeeping, included injections and heats, relevant processing and attributes, and tandem segment layout. Changing a result name or attaching a different solution to an experiment does not by itself change this status. Validity does not say whether the model is scientifically appropriate or the fit is good.

- **Analysis is valid** means the saved fit inputs still match.
- **Partially invalid** means some members of a multiple-experiment result fitted independently without temperature dependence have changed, while others still match. A shared-parameter fit or a result with temperature dependence is treated as invalid when any member's fit inputs change, including when its members were fitted individually.
- **Invalid** means the saved fit no longer matches the current inputs, or the set of member experiments has changed.
- **Unknown status** means FT-ITC cannot make the comparison, for example because an older result has no validity snapshot.

Check the listed validity reasons and use **Update Result** when inputs have changed. Stored fitted parameters continue to describe the earlier inputs until updated.

Removing any source experiment from the experiment browser makes the whole result **Invalid** and names the missing experiment. **Undo** restores the previous validity when it restores the original experiment unchanged.

Analysis warnings are separate from validity: a matching fit can still reach a parameter bound, encounter an uncertainty limit, or have an **Inconclusive** member. A manual **Binding detected** conclusion clears only that member's Inconclusive warning. No-binding members' boundary and uncertainty-limit warnings remain listed with the experiment but do not contribute to result status. Repeated desktop reasons are summarized by category and count; the browser likewise distinguishes saved analysis warnings from input validity.

Information criteria

AIC and **AICc** help compare candidate models fitted to the same observations. They balance agreement with the data against the number of fitted parameters; a smaller value is preferred within a comparable set. They do not tell you that any model is scientifically correct.

Read **Information criteria** for the analysis-level AICc, or AIC when its small-sample correction is unavailable, plus observation and parameter counts. AICc requires $n > K + 1$. If the likelihood is unavailable, inspect the displayed reason.

For independent fits, the **AICc / AIC** column compares alternative models for the same experiment, observations, response definition, and weighting mode. Do not compare scores across different experiment rows. Shared-parameter fits show only the analysis-level criterion. Prefer AICc when available; neither criterion establishes model adequacy.

Likelihood calculation

The criteria use the saved result's included injections and response definition. At analysis level, residual statistics are pooled before estimating the variance, including when members were fitted independently. Unweighted criteria estimate one common variance across all members; weighted criteria estimate one common multiplier of the injection-error variances. Each independent member's own criterion estimates its variance separately. Consequently, neither pooled AIC nor pooled AICc is a sum of the member values.

With residuals r_i, raw residual sum of squares $\mathrm{RSS} = \sum r_i^2$, integration-error SDs σ_i, and standardized residual sum of squares $Q = \sum (r_i/\sigma_i)^2$, the maximized likelihood terms are:

For an **unweighted fit with one estimated common variance**:

CALCULATION:

$$-2\log L = n[\log(2\pi \mathrm{RSS}/n) + 1]$$

This estimates one shared residual variance from the total squared mismatch RSS, then scores how likely the observed heats are under that estimate.

For a **weighted fit with one estimated variance multiplier**:

CALCULATION:

$$-2\log L = n[\log(2\pi Q/n) + 1] + \sum \log(\sigma_i^2)$$

This uses each injection's processing SD σ as a relative weight. The score also includes their different uncertainty scales.

Weighted criteria use the processing SDs as relative measures of injection uncertainty and estimate one common variance multiplier, Q/n, from the standardized residual score Q and observation count n. This does not change those SDs or the fitted parameters. Every included injection needs a positive SD for weighted fitting. A score is unavailable when residual variance is zero. Weighted profile-likelihood intervals retain the relative weights and use the estimated overall residual scale for their threshold.

The fitted parameter count p includes only parameters free in the saved global model. Shared coordinates count once; member-specific coordinates count once per member. For a member criterion, p includes only that member model's free fitted parameters. In both weighting modes,

the reported values use $K = p + 1$, $\mathrm{AIC} = -2\log L + 2K$, and $\mathrm{AICc} = \mathrm{AIC} + 2K(K+1)/(n-K-1)$. This standard small-sample correction is an approximation for nonlinear ITC models.

Binding assessment

Review the automatic conclusion beside the fit and residuals. **No binding detected** means this experiment does not establish binding relative to the Null model; it does not establish that the molecules cannot bind. Use **Modify Assessment** to override the conclusion or restore automatic assessment. Updates recalculate the automatic outcome and retain manual overrides; reopening applies the current rule to saved comparison evidence without refitting. Assessments are saved in `.ftxtc` projects.

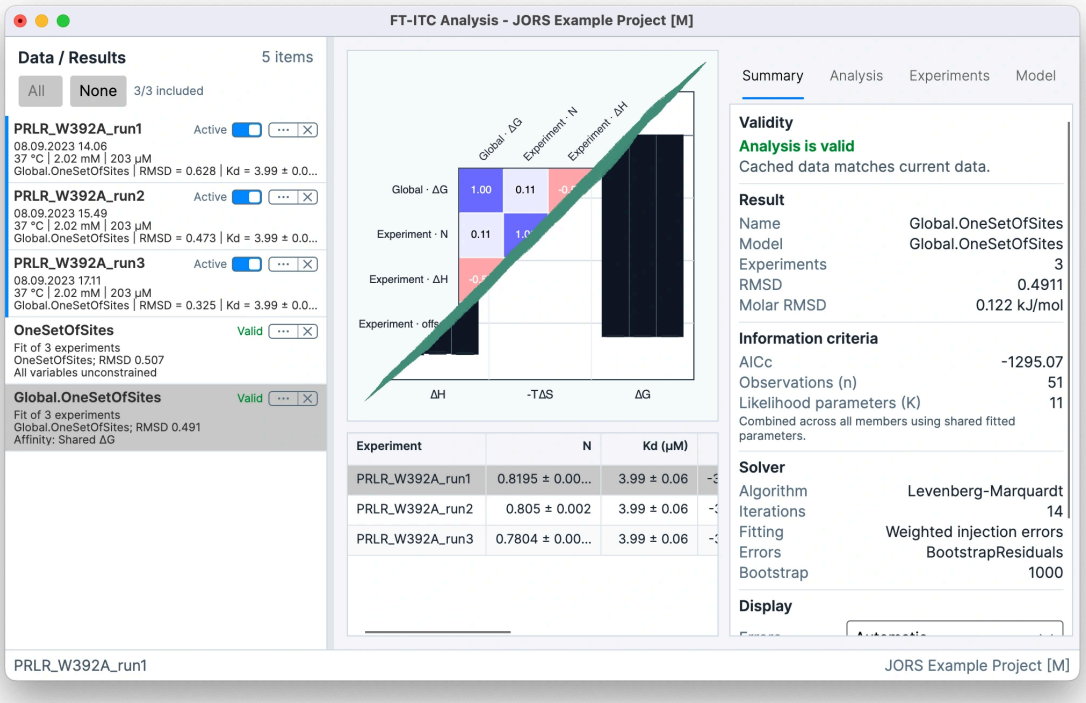
Null prediction overlays the model fit on the integrated-heats graph; the original binding fit remains available. In the native macOS Analysis view, choose **Show Null Prediction** from the **Analysis** menu on the right side of the toolbar.

Uncertainty estimation (bootstrap, leave-one-out or profile likelihood) is not run for a fit assessed as **No binding detected**, because its parameters are not expected to be meaningful; its uncertainty is shown as **Not estimated (no binding detected)**. In an independent collection this applies to each experiment separately, and combined uncertainty is then unavailable. Manual assessments count: if you mark such a fit **Binding detected**, the result is marked stale until **Update Result** estimates its uncertainty.

Independent experiments have individual assessments. **Member assessments** shows a common conclusion or **Mixed assessments**, including manual overrides; it is not a joint statistical verdict. Inconclusive members retain binding values and uncertainty. **Not assessed** imposes no restriction and is not positive evidence. If any member is effectively **No binding detected**, combined binding values and dependent analyses are omitted, with those members named; values are never recomputed from a subset. A pooled comparison is diagnostic only. A shared global fit instead has one result-level assessment.

Standard reports retain every member and its model, options, scope, constraints, fixed settings, and fit details. No-binding parameter and RMSD cells are blank; their parameter tables, c-values, correlations, and fit warnings are omitted. **No binding detected** and **Inconclusive** appear under the experiment name. Unavailable estimates use **—**, distinct from suppressed blank cells. Standard no-binding figures show the Null prediction against buffer-corrected observations without fitted offset correction. Manual conclusions are marked **(manual)**, including collection summaries with an overridden member. Thermodynamic summary plots omit no-binding members in both Standard and Diagnostic reports, name the omissions, and disappear if none remain. Inconclusive members retain binding output.

Standard result-table exports continue to omit no-binding members' binding values and RMSD, while keeping their ionic strength and ionization enthalpy; Inconclusive values are included. **Diagnostic report** and **Diagnostic export** retain available attempted binding detail, including every member's estimates, labels, and fit warnings, and full comparison evidence. Diagnostic reports place pooled comparison evidence for independent collections in **Pooled Comparison Diagnostics**; Diagnostic Summary exports place it in separately labelled pooled-diagnostic columns. This evidence uses a pooled variance convention and does not determine member assessments.



Null comparison calculation

A null hypothesis is the simplest explanation of the data that you want to rule out before trusting a more detailed model. For a titration, the null hypothesis is that the measured heats can be explained without binding. A Null model represents that no-binding explanation, including heat from dilution and mixing. If a binding model does not describe the data clearly better than this background alone, its fitted affinity, enthalpy, and stoichiometry are not supported by the data.

The comparison uses the Offset Null model, which predicts each injection heat as $q_i = bm_i$, where m_i is the amount injected and b is a constant background heat per mole of injectant. It uses the same included injections and weighting as the binding fit. The analysis inspector shows the automatic conclusion; use **Modify Assessment** on the saved result to change it manually or return to automatic assessment. The Summary shows the Null model RMSD and the signed $\Delta\mathrm{AICc}$. For multi-experiment results, the Null model is fitted separately for each experiment using the ordinary single-experiment fit. A locked or shared binding-model does not constrain these local Null model fits. For independently fitted binding models, each local comparison determines its member assessment. For pooled shared-parameter binding fits, all local null fits are combined under the pooled variance convention to calculate the result-level comparison.

The two fits are compared with AICc (see Information criteria), which balances fit quality against the number of fitted parameters:

CALCULATION:

$$\Delta\mathrm{AICc} = \mathrm{AICc}_{\mathrm{null}} - \mathrm{AICc}_{\mathrm{binding}}$$

A positive value favors binding because the Null model has the higher AICc. It does not establish that the selected binding model is adequate or correct. A negative $\Delta\mathrm{AICc}$ value indicates that the Null model is favored.

The chosen cutoffs are **No binding detected** at $\Delta\mathrm{AICc} \leq 0$, **Inconclusive** at $0 < \Delta\mathrm{AICc} < 15$, and **Binding detected** at $\Delta\mathrm{AICc} \geq 15$. They have no calibrated false-positive guarantee; the comparison is not a significance test.

Uncertainty and evaluation temperature

The central value is the model's best fit to the original data. An uncertainty display gives a range or spread around that value under the selected estimation method. It does not include every possible experimental error or guarantee that the chosen model is correct.

The **Errors** display control provides **Automatic**, **Standard deviation**, **95% confidence interval**, and **SD + 95% CI**. **Standard deviation** presents the primary best-fit value with a symmetric ± SD; **95% confidence interval** presents that same best-fit value with its lower and upper confidence limits; and **SD + 95% CI** presents both. The central value is always the primary best fit, not the mean or median of the resampled values. Residual-bootstrap intervals use percentile limits; profile-likelihood intervals use the likelihood-threshold endpoints, with an equivalent symmetric scale for the SD display.

The **Parameter Evaluation** section contains an evaluation **Temperature** field and the displayed thermodynamic quantities at that temperature. Its default is the current **Reference temperature** when the measured temperature span exceeds the configured minimum, including independently fitted series; otherwise it is the mean experiment target temperature. Temperature display can be **Celsius** or **Kelvin**. Results remembers this unit for the current application session, independently of the result exporter. Both start from **Preferences > Temperature units**; changing that preference resets both choices. Switching units preserves the physical evaluation temperature. Changing the evaluation temperature updates the displayed quantities calculated from the stored model; it does not change injection heats or refit the result. Temperature-dependent values are meaningful together with their model, units, uncertainty representation, and evaluation temperature.

Changing **Errors** changes only how stored uncertainty is presented in tables, parameter evaluation, and graphs. It does not rerun the fit, change the best-fit parameters, or turn one error-estimation method into another. Bootstrap construction, parameter transformation, and the SD and percentile calculations are described under Parameter uncertainty.

Automatic display calculation

Automatic shows the 95% confidence interval when the interval is clearly lopsided around the best-fit value, and SD otherwise. In practice, CI is shown when the longer side of the interval is at least about 1.44 times the shorter side. A symmetric `value ± SD` would misrepresent such an interval, so its limits are shown instead.

The choice is made separately for each reported quantity. Let L and U be its stored 95% confidence limits and $\hat{\theta}$ its primary best-fit value:

CALCULATION:

$$w_{\text{lower}} = \hat{\theta} - L; \quad w_{\text{upper}} = U - \hat{\theta}$$

$$\text{asymmetry} = \frac{|w_{\text{upper}} - w_{\text{lower}}|}{w_{\text{upper}} + w_{\text{lower}}}$$

Automatic shows the 95% confidence interval when both widths are positive and the asymmetry is at least 0.18; otherwise it shows SD. An asymmetry of 0.18 corresponds to one side being 1.18/0.82 ≈ 1.44 times the other. This is a display rule based on interval imbalance around the best fit, not a formal statistical test of distribution skewness.

For example, K_d = 50 nM with a 95% interval of 40–70 nM has widths 10 and 20 nM. The asymmetry is 10/30 = 0.33, so Automatic shows the interval. With an interval of 42–59 nM, the widths are 8 and 9 nM, the asymmetry is 1/17 = 0.06, and Automatic shows SD.

To check a value, set **Errors** to **SD + 95% CI**. Both the SD and the limits are then shown, and the widths can be read directly. If a quantity has no stored interval, Automatic shows SD.

The decision is made after a fitted coordinate has been transformed into the displayed quantity. A nonlinear transformation—such as fitting $\log_{10}(K_a)$ and displaying K_d—can therefore make the displayed interval asymmetric and cause **Automatic** to select CI for that quantity.

Summary uncertainty

Use **Combined SD** to describe combined spread and the **Approximate propagated interval** for the average or evaluated temperature trend. Neither replaces individual parameter uncertainty.

For local parameters, **Combined SD** is the square root of between-experiment sample variance plus mean individual variance; a temperature trend uses residual variance. It describes combined spread, not the standard error of the mean, and observed spread may already include fitting noise.

The **Approximate propagated interval** describes the average or evaluated trend. It combines observed spread with separately propagated lower and upper individual interval widths, preserving possible asymmetry. Two distinct-temperature observations propagate individual errors without estimating residual spread; additional observations contribute residual spread. ΔC_p uses slope uncertainty.

These intervals lack established 95% coverage, especially for small collections, and omit between-experiment covariance even when local parameters are coupled through a global fit. Shared and temperature-constrained parameters retain model-estimated uncertainty. Central values remain

the fits to the original data.

Summary K_d is calculated from summarized Gibbs energy. Its SD is propagated from Gibbs-energy SD using the local slope of the exponential transformation, so it is an approximation; its interval endpoints are transformed from the Gibbs-energy interval. Individual-fit uncertainty retains its own calculation.

Parameter correlation

This view shows whether two fitted parameters tend to move together when the data are resampled. A strong relationship can mean that the data have difficulty separating their effects, but the display alone cannot establish why.

Correlation shows Pearson correlations between fitted parameter coordinates across residual-bootstrap refits. It requires **Bootstrap residuals**, at least 30 complete refits, and at least two parameters that vary across those refits. Parameters with no variation are omitted. Parameters fixed in the primary fit are also omitted unless **Unlock parameters** allowed them to vary during bootstrap error estimation; such parameters are marked with an asterisk and a warning that bootstrap parameter unlocking was enabled.

Matrix values range from –1 to +1. Point to a cell for Pearson's r and its Monte Carlo precision interval. An interval spanning zero leaves the sign unresolved at the current simulation precision. Check the panel's refit counts and warnings for coarse precision, failed or incomplete refits, and rank limits.

For independent fits, select a member; correlations, intervals, counts, and warnings use only its refits. A joint fit can show **Shared** coordinates or **Shared + selected local** coordinates. The labels identify each parameter's scope.

Correlation calculation and limits

CALCULATION:

$$r_{jk} = \frac{\sum_b (\theta_{bj} - \bar{\theta}_j)(\theta_{bk} - \bar{\theta}_k)}{\sqrt{\sum_b (\theta_{bj} - \bar{\theta}_j)^2 \sum_b (\theta_{bk} - \bar{\theta}_k)^2}}$$

r_{jk} is the Pearson correlation between fitted coordinates j and k. The index b runs over complete residual-bootstrap refits, and $\bar{\theta}$ is the corresponding mean coordinate.

For each off-diagonal cell, the application also reports an approximate 95% Monte Carlo precision interval using Fisher's transformation:

CALCULATION:

$$z = \operatorname{atanh}(r), \quad z_{\pm} = z \pm \frac{1.959964}{\sqrt{B-3}}, \quad r_{\pm} = \tanh(z_{\pm})$$

B is the number of complete refits: usable refits with finite values for every displayed coordinate. The diagonal is the structural self-correlation $r = 1$ and has no Fisher precision interval.

Sequential correlations use the actual fitted coordinates. Affinity coordinates are labeled **log10 Ka1** through **log10 Ka4**, active enthalpy coordinates are included, and no N-value coordinate is added. A global result shows the shared per-step coordinates and, when requested, unconstrained coordinates for the selected member without duplicating constrained member values.

INTERPRETATION:

Correlation shows how fitted coordinates varied together under the residual bootstrap. The Fisher interval describes the finite-bootstrap Monte Carlo precision of the correlation coefficient r itself; it is not a confidence interval for either fitted parameter and does not replace the parameter uncertainty display. It does not establish identifiability, causality, model adequacy, or model validity. Frequent refit failures can make the retained ensemble selective, and the Fisher interval does not account for those failures. Affinity is evaluated in the fitted coordinate system—$\log_{10}(K_a)$—rather than as the displayed K_d. A rank warning states the structural limit $\text{rank} \le B - 1$ for the displayed parameter count; it is distinct from numerical rank and scientific model validity.

Advanced analysis views

These views ask how a fitted interaction changes with experimental conditions. **Temperature** examines a series measured at different temperatures, **Salt** examines a series with different salt conditions, and **Protonation** compares experiments in different buffers. Their conclusions depend on the recorded conditions and the underlying fit.

All advanced analyses require a **One-Set-Of-Sites** Analysis Result. Each analysis also requires the relevant variation in experimental conditions and the corresponding metadata. The advanced analyses operate on the stored member solutions and expose their own calculated outputs; they do not change the base fit parameters. A sequential result can still show its ordinary per-step ΔH, ΔG, $-T\Delta S$, K_d, and temperature-dependence presentation; it reports structuring, protonation, and electrostatics as unsupported by that model rather than hiding the ordinary thermodynamic views.

Temperature

The **Temperature** view is available when the difference between the highest and lowest member temperatures exceeds the configured minimum span. Its **Structuring** section uses the Spolar–Record method to estimate the hydration entropy ($-T\Delta S_{HE}$), the conformational entropy ($-T\Delta S_{conf}$), and the number of residues folding upon binding. **Interaction** selects **Folded–folded** for two folded proteins or **Folded–disordered** for one folded and one disordered protein. Completed outputs retain their saved estimates and evaluation temperature until rerun.

Structuring assumptions

Structuring analysis is evaluated at the **Isoentropic point**, where the model-estimated binding entropy is zero ($\Delta S = 0$), following the published method. These are model-based estimates, not direct structural measurements. For a thermodynamically linked result, the isoentropic temperature is derived from the complete Gibbs, enthalpy, and heat-capacity relationship. Profile intervals are propagated approximately and omit fitted-coordinate correlations; bootstrap and leave-one-out uncertainty evaluates each saved relationship jointly, while the Spolar–Record Monte Carlo samples its linked inputs independently.

Salt

The **Salt** view requires a **Salt** attribute for every member and at least three ionic-strength conditions with usable affinities. Select conditions in increasing order, each more than half the **Minimum ionic-strength span (mM)** preference above the previous selected condition. Its **Graph mode** values are **Affinity vs Salt**, **Debye-Huckel**, and **Salt Dependence**.

Salt-dependence slope describes how the fitted dissociation constant, Kd, changes with recorded salt concentration. Use the same single salt throughout the collection, with at least three distinct positive concentrations and usable affinities. Repeated entries of that salt are added together, and repeated experiments remain separate observations. Zero-concentration controls are excluded from logarithmic fitting. Mixed salts or changing salt identities make this slope unavailable; the separate ionic-strength analysis may still be available.

The **Salt Dependence** graph identifies the salt and plots `ln(Kd / M)` against `ln(Salt concentration / M)`:

$$\ln(K_\mathrm{d}/1\ \mathrm{M}) = b + s\ln(c_\mathrm{salt}/1\ \mathrm{M}).$$

Here, s is the **Salt-dependence slope**, b is the fitted intercept, and c_salt is the recorded salt concentration. A positive slope means salt weakens binding; a negative slope means salt strengthens binding. For example, a slope of +2 means doubling salt concentration increases Kd fourfold. The same definition applies to monovalent and multivalent salts.

This calculation uses concentration without activity corrections or inferred ions from buffers and pH adjustment. Ion release or uptake is a possible interpretation under an appropriate binding model; the slope alone does not establish an ion count. Keep pH, temperature and other conditions comparable when interpreting a salt series. The distinction between salt activity and mean ionic activity follows IUPAC, Green Book, §4.13; the model dependence of ion-release interpretations is discussed by Record et al. (1976) and Chowdhury et al. (2023), Eq. 3.

Kd0 is the extrapolated dissociation constant at zero ionic strength from the separate ionic-strength fit:

$$\ln K_\mathrm{d}(I) = \ln K_{\mathrm{d},0} + s_I\sqrt{\bar{I}},$$

where I is ionic strength and s_I is that fit's salt sensitivity. Saved ionic-strength plots retain the buffer-inclusion setting used for their fit. A new calculation uses the setting captured when the result was opened or updated.

After changing inputs used by either salt analysis, use **Run Analysis** again to obtain updated results. Affected outputs remain unavailable until recalculated; the other component remains available if its inputs are unchanged.

Each point represents one experiment's fitted affinity and complete uncertainty interval, transformed to the graph's scale. The reported slope is the regression of the original fitted affinities; random input sampling estimates its uncertainty. Recalculate older counter-ion results to obtain the concentration-based slope; their primary fits and saved ionic-strength results remain available.

Protonation

The **Protonation** view requires a **Buffer** attribute and available buffer ionization enthalpy at the experimental temperature for every member, with at least two distinct enthalpy values across the collection. **Available analyses** explains which buffers and experiments lack usable thermochemistry. These buffers remain selectable for ordinary fitting; missing thermochemistry is unavailable rather than zero, and members are not automatically omitted to enable protonation analysis. Its calculated output contains **Net proton uptake** and **Buffer-independent enthalpy**, with uncertainty when the stored analysis contains the corresponding uncertainty results.

The view relates the stored member binding enthalpies to the buffer ionization information associated with their conditions. The result is conditional on the recorded buffer identities, their temperature-dependent ionization enthalpies, and the model assumptions; it does not identify a particular residue or microscopic protonation event.

The band spans lines combining the ends of the slope's and intercept's 95% intervals, ignoring their correlation; it is neither a prediction interval nor a joint confidence band, and extrapolating the protonation intercept to zero buffer enthalpy can make it wide.

CALCULATION:

$$\Delta H_{\text{obs}} = \Delta H_0 + \Delta n_{\text{H}^+}\ \Delta H_{\text{ion,buffer}}$$

Net proton uptake, Δn_{H^+}, is the fitted slope: positive means uptake by the binding system from solution, and negative means release. It is an average net exchange per binding event and may be fractional. The x-axis is buffer ionization (deprotonation) enthalpy; buffer protonation enthalpy has the opposite sign. **Buffer-independent enthalpy**, ΔH_0, is the fitted intercept at zero buffer ionization enthalpy, at the experimental pH. It retains contributions from protonation of the binding partners and is not the fully intrinsic binding enthalpy. Interpret the relationship for comparable conditions, including pH and temperature. See Coussens, Schuck and Zhao (2012), equations 3–6 and Goldberg, Kishore and Lennen (2002).

Advanced-analysis values are supplemental views of a stored Analysis Result. Their availability and outputs are determined by the One-Set-Of-Sites model, member variation, metadata, selected graph or evaluation mode, and any completed uncertainty calculation. Result validity remains a separate indication of whether the stored fit inputs match the current project state. Figure and table output is covered in Figures and export.

The buffer registry contains approximations and some unknown quantities, so confirm that the chosen buffer values apply to the experiment. The TAPSO pKa temperature correction is a local approximation for 20–30 °C. Its protonated zwitterion is neutral, and imidazole's ionization heat-capacity slope is –9 J/(mol K). Recalculate affected derived analyses when comparing with results produced before these corrections.

When **Include buffer in ionic-strength calculation** is enabled, each buffer is corrected using only its own contribution and the transition whose pKa is nearest the recorded pH. The estimate treats that transition as two species and assumes monovalent counterions from acid/base adjustment; salt and other buffer contributions are added separately. This is an approximation, and numerical convergence does not establish chemical accuracy at high buffer concentrations.

CHAPTER 09

Figures and export

Configure final and supporting figures, print active graphs, and export data, results, figures, and complete analysis reports.

Last verified: 2026-09-23

Choose a figure when you want to show a graph, a data export when you need injection values, and a result table or report when you need to share fitted quantities and their context. Check the selected experiments, units, and uncertainty display before sending an output.

FT-ITC Analysis keeps publication figures, numerical data, and fitted result tables as separate output types. **Final Figure** is an Experiment Data workflow. An Analysis Result does not open directly as a Final Figure; **Export Associated Final Figures...** on a result exports figures for the experiment solutions associated with that result.

Choose the appropriate output tool

The Analysis Report is distinct from the other output tools:

- **Analysis Result Exporter** writes compact CSV or TSV tables for numerical reuse.
- **Final Figure** writes a publication figure for an experiment, while **Export Associated Final Figures...** writes one figure per result member.
- **Supporting Figure** composes selected figures into one multi-panel canvas.
- **Analysis Report** assembles one or more saved results as self-contained chapters with figures, detailed tables, diagnostics, optional advanced plots, and source and analysis history.

Final Figure

The **Final Figure** workspace renders the selected Experiment Data as a publication figure. Its preview and PDF output reflect the selected experiment, processing state, fitted solution, and figure options. To export figures using the fits stored in an Analysis Result, use **Export Associated Final Figures...** on that result.

The **Export PDF** controls offer three scopes:

- **Selected** opens a save dialog for the displayed experiment figure and suggests the experiment name as the PDF filename.
- **Active** exports figures for Active Experiment Data. **All** exports figures for all Experiment Data in the project.

For **Active** and **All**, choose a parent folder and the app creates or reuses a subfolder named after the saved project file (without its extension). For an unsaved session, it uses the local export date (`yyyyMMdd`) as the folder name.

The result-list command **Export Associated Final Figures...** writes one final-figure PDF per member experiment using that result's saved fits and the current **Plotted model** choice. Choose a parent folder; the app creates or reuses a subfolder named after the result. It is available for a result with exportable solutions. See Results and advanced analyses for result validity and stored member-solution behavior.

All figure batches use experiment names for their PDF filenames. When names in the same batch would collide, numeric suffixes such as `Experiment (2).pdf` keep them distinct. If any destination PDFs already exist, the app asks once before replacing them; **Cancel** stops the entire export without writing files. Repeating an export replaces the approved PDFs rather than creating additional suffixed copies.

Final Figure preview, PDF export, and printing use the same selected model. Associated and batch exports also use this choice.

WHERE FINAL FIGURE SETTINGS ARE USED

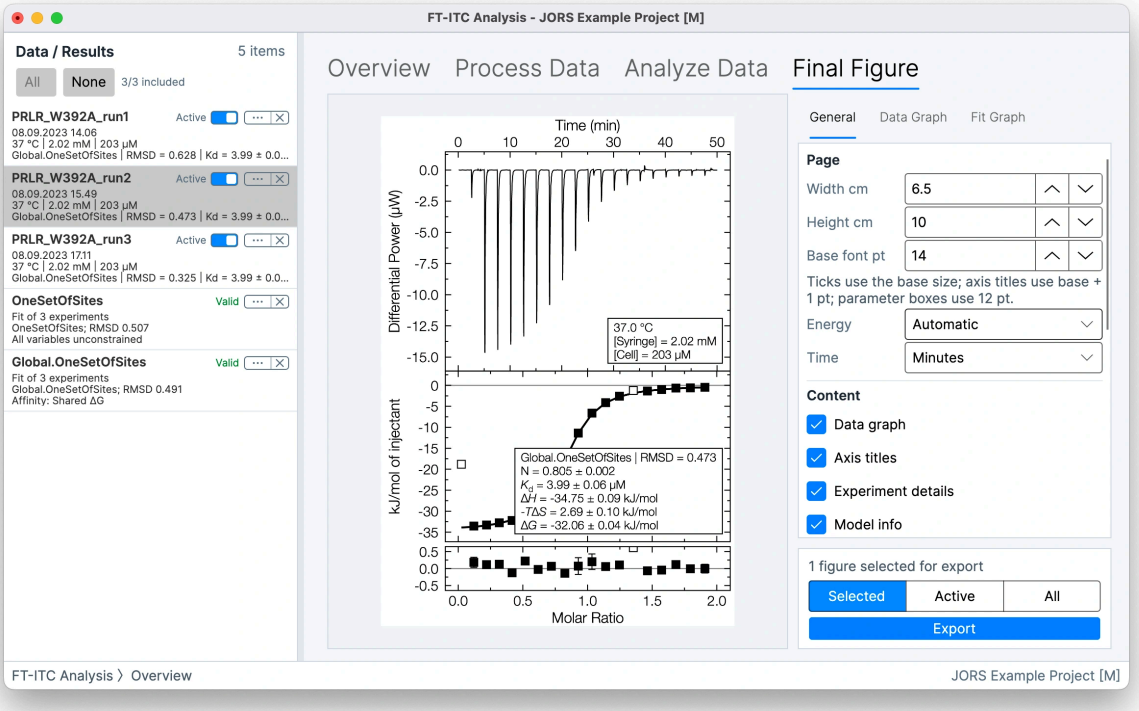
- **Final Figure PDFs**, including Active, All, and **Export Associated Final Figures...**, use the current Final Figure appearance controls.
- **Supporting Figure...** starts with a snapshot of the appearance controls when its window opens. This includes units, axis limits, visible traces and fits, and excluded-point automatic Y scaling. Its model selection follows the existing assessment policy, independently of **Plotted model**. Its own plot size, grid, typography, panel labels, grouping, and information-box controls take precedence; later Final Figure edits do not update an already open Supporting Figure.
- **Analysis Report** uses a separate compact figure preset and its own energy and uncertainty selections. Final Figure axis limits and excluded-point automatic Y scaling do not carry over to report figures.

General

The **General** tab defines the page and common content. Page controls specify plot-area width and height in centimeters and the base font size. **Energy units** selects **Joule** or **Calories** for the current app session, starting from the Preferences energy units. The base or kilo prefix is chosen from the figure's plotted and reported central values; with a shared enthalpy axis, all experiments use one unit. The selected family also controls thermogram units: differential power is **µW** or **µcal/s**, while integrated heat is **µJ** or **µcal**. Time controls set the displayed time unit. Content controls include the data graph, axis titles, experiment details, model information, fit parameters, and the information-box placement. The uncertainty selector provides **Automatic**, **SD**, **CI**, **SD + CI**, and **None**. **Automatic** uses the same per-quantity asymmetry rule described under Uncertainty and evaluation temperature.

The parameter controls determine which information appears in the information box: thermodynamic, derived, or offset parameters; temperature; concentrations; injection delay; instrument; and user-defined attributes. The information box is descriptive figure content and does

not alter the underlying fit.



Data Graph

The **Data Graph** tab controls the differential-power trace. Power and time axis titles, tick density, and explicit minimum and maximum values define the axes. **Corrected data** selects the baseline-corrected trace when one is available. **Shared power axis** applies one power-axis range across the active experiments represented by a figure set.

Baseline controls expose the baseline, its **Solid** or **Dashed** style, **Under data** or **Over data** layer, and line width. **Integration ranges** displays the integration intervals as **Bar**, **Fill**, or **Endpoint lines**. These overlays describe processing and do not recalculate integration.

Fit Graph

The **Fit Graph** tab controls the integrated-heats and model panels. Enthalpy and molar-ratio axis titles, tick density, and explicit ranges define the axes. Symbols are **Square** or **Circle**, with an independent point size. **Shared X axis** and **Shared enthalpy axis** unify the corresponding ranges across active experiments; the residual-axis range follows the shared enthalpy setting when configured for unified residual axes.

Model groups the plotted model, line width, smoothness, confidence band, and offset correction. **Plotted model** selects **Binding**, **Null**, or **None**. It starts at Binding and keeps your choice across experiments for the current app session. The curve, residuals, fitted parameters, and automatic units and axes follow that choice. The saved assessment, or **No assessment**, is shown below the control; selecting a model neither changes the assessment nor refits the data. You can display the binding model even when the assessment is **No binding detected**.

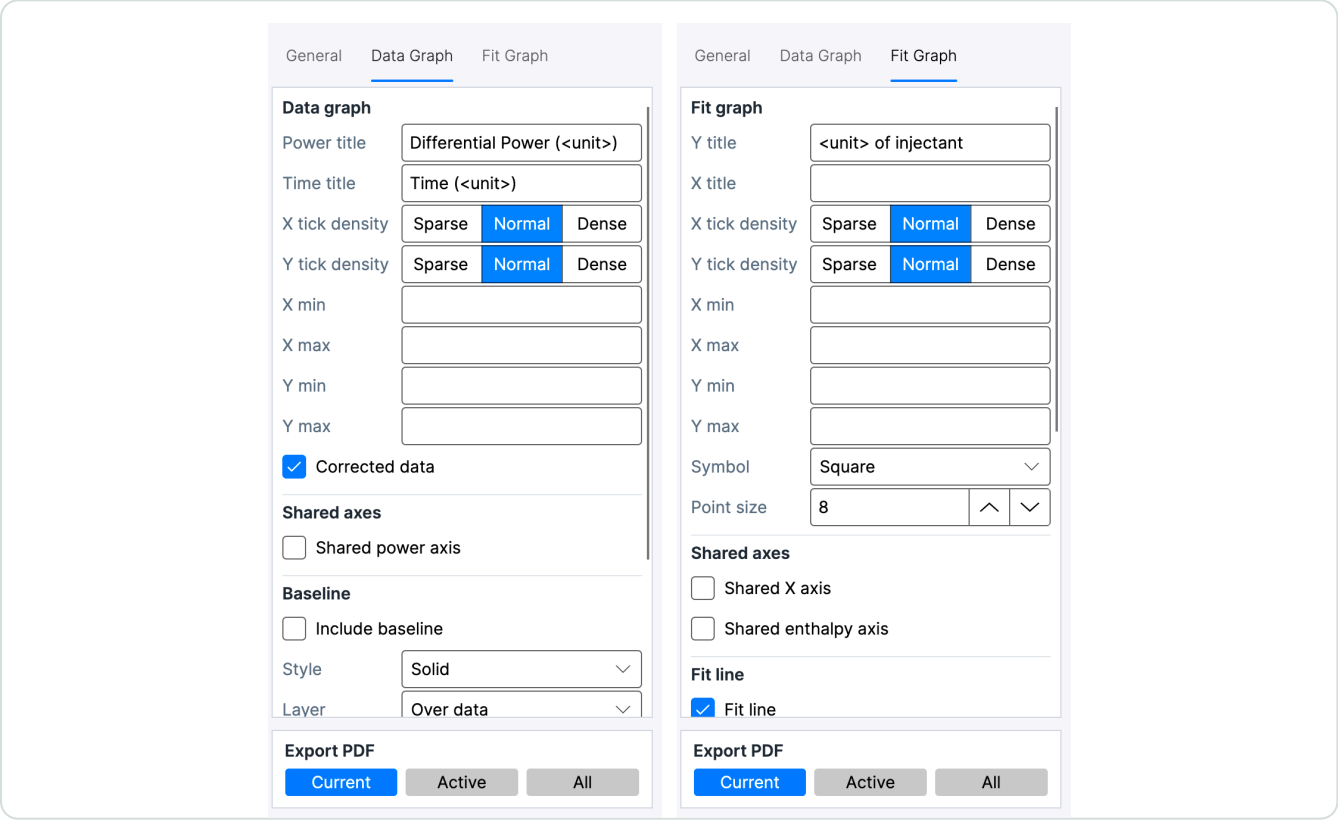
Null model figures show the full saved dilution-heat prediction against buffer-corrected observations. Offset correction and confidence bands are disabled while Null is selected; their settings are retained when you return to Binding.

Auto Y axis includes excluded points lets excluded injections affect the automatic integrated-heat and residual axis ranges. It starts from the application preference. Explicit axis limits take precedence.

None shows processed observations without a model curve, confidence band, residuals, fitted parameters, model information, parameter guides, or fitted offset correction. Inactive controls keep their settings for when you return to a compatible model.

Line width sets the model curve thickness. **Smoothness** selects **Smooth**, **Spline**, or **Linear** interpolation. **Show residuals graph** adds the residual panel, and **Residual gap** separates it visually from the fit panel. The display controls include the zero enthalpy line, error bars, excluded points, and excluded error bars. **Confidence band** and **Offset correction** are in **Model**. Error bars use processed injection uncertainties; confidence bands require suitable stored resampling solutions and are unavailable for profile likelihood; fit lines and residuals require a fitted solution.

Traceability Mode keeps excluded injections visible in figures and disables the excluded-point visibility control. This does not include those injections in the fit.



Numerical data export

File > Export Data... opens the numerical export dialog. **Export Selected Data...** invokes the same export for the selected experiment. The data scope is one of **Selected experiment**, **Active experiments**, or **All experiments**.

The format list contains:

- **Thermogram Data** — a `.csv` file containing time and power samples, with the **Export baseline-corrected trace** option when baseline-corrected samples exist.
- **Integrated Peaks** — a `.csv` file containing injection-axis values, molar heats, their SD, fitted values, and residuals when available, with **Export offset-corrected peaks** when a fitted solution is available.
- **Combined Data** — a `.csv` file placing thermogram samples and integrated-peak columns side by side. It includes raw and baseline-corrected traces when available; peaks and fitted values are exported without fitted-offset correction, with residuals when available.
- **MicroCal / SEDPHAT** — a `.dat` file with MicroCal-style `DH`, `INJV`, `Xt`, `Mt`, `XMt`, `NDH`, `DY`, and `Fit` columns. `DH` is in µcal, `INJV` is in µL, concentrations are in mM, and `NDH`, `DY`, and `Fit` are in cal/mol. The first injection uses `--` for normalized heat and residual, following the conventional excluded-first-injection layout.
- **pytc** — a `.dh` file containing the pytc-compatible injection and metadata fields.
- **ITCsim** — a `.csv` file containing ITCsim-compatible injection data and metadata, with offset-corrected peaks available when a fitted solution exists.

Molar-energy unit is available for Integrated Peaks, Combined Data, and ITCsim: choose **J/mol**, **kJ/mol**, **cal/mol**, or **kcal/mol**. Each export dialog starts at kJ/mol for the Joules energy preference or kcal/mol for the Calories preference; a choice applies to that export only. Integrated Peaks and Combined Data identify the chosen unit in the heat, SD, model, and residual column headers. ITCsim keeps its `PeakHeat` column name and records the unit as `#EXPINFO ENERGYUNIT J`, `KJ`, `CAL`, or `KCAL`; those tokens denote energy per mole. Peak area is an absolute heat in J and is distinct from the exported molar heat.

Numerical exports use a period as the decimal separator, regardless of regional settings.

Other output units remain format-specific. Thermogram samples use seconds and watts; MicroCal/SEDPHAT and pytc use their documented concentration, volume, temperature, and heat conventions. Fitted columns and correction controls are disabled when the selected data do not contain the corresponding processed or fitted state. Export defaults are described in Settings and defaults.

These exports are not project backups. General `.csv` and `.tsv` exports cannot be reopened through **File > Open...**. A MicroCal / SEDPHAT `.dat` export or pytc `.dh` export can be reopened as integrated-heat input, but doing so restores neither a thermogram nor the FT-ITC project state and requires choosing the source heat unit. Select **microcalorie** for the `DH` column when reopening a newly generated MicroCal / SEDPHAT export. Historical FT-ITC `.dat` exports may instead require **joule**, matching the unit used when they were created. Save an `.ftxtc` project when the analysis must remain editable.

Analysis Result Exporter

Copy Result Table copies one row per experiment as tab-separated text for pasting into a spreadsheet. Values and uncertainties occupy separate columns, following the uncertainty display preference; **None** copies values only. For combined summary rows or other table formats, use **Analysis Result Exporter...**.

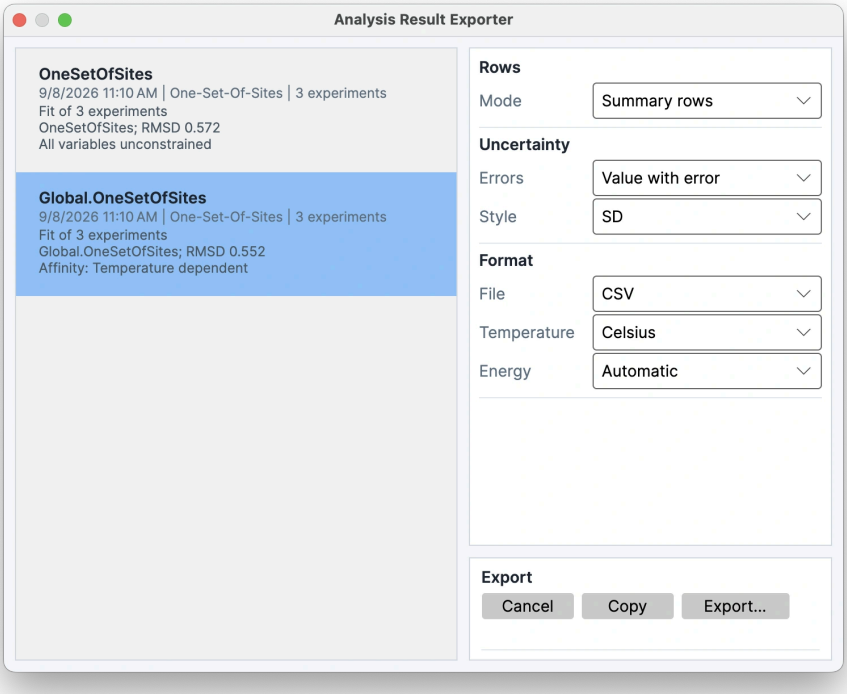
Analysis Result Exporter... builds a table from one or more selected Analysis Results. **Summary rows** produces one row per result; **All replicate rows** produces one row per member fit. Error layout is **Value with error** or **Separate columns**. Uncertainty style is **SD**, **CI**, or **SD + CI**. The file format is **CSV** or **TSV**. **Energy units** provides **Automatic**, **J**, **kJ**, **cal**, and **kcal**. Automatic chooses one stable molar-energy unit across all selected results and one independent unit for ΔC_p columns; a fixed choice uses the selected energy unit consistently. Temperature presentation is **Celsius** or **Kelvin**; changing the unit preserves the physical evaluation temperature. The exporter remembers its unit when reopened during the current application session, independently of Results. Both start from **Preferences > Temperature units**; changing that preference resets both choices.

The configured table is available through **Copy** and **Export...**. Copy places the same delimited text on the clipboard; Export writes it to a file. Numeric values in exported and copied result tables use a decimal point regardless of regional settings. Profile-likelihood SD columns use the equivalent display scale implied by their stored endpoints and are labeled as such in exported metadata; the exact lower/upper endpoints remain available in separate-column mode. The exporter does not include the parameter-correlation matrix, which is calculated for display rather than stored as a result-table field.

Summary rows use the same Parameter Evaluation calculation at its default evaluation temperature: the configured reference temperature for a temperature series, otherwise the mean experiment temperature. The exported temperature column labels that evaluation temperature. Local summary rows report **Combined SD** and an **Approximate propagated interval**, using the same calculation described under Parameter Evaluation. Summary interval columns have neutral `_interval_lower` and `_interval_upper` suffixes; individual rows retain their CI95 columns. K_d is derived from the summarized Gibbs energy rather than averaged directly, and ΔC_p columns are included for applicable temperature-dependent results. No results are pooled across separate analyses.

Result tables, clipboard output, final-figure metadata, and viewer output create thermodynamic columns according to the fitted model. A sequential result therefore exports one K_d, ΔH, ΔG, and $-T\Delta S$ value for every active step, including steps 3 and 4 when present. The fixed step count is included as model information. Exporting preserves the interpretation of these values: macroscopic sequential constants remain step constants, not microscopic site constants. Fitted N, α and fixed stoichiometry occupy separate columns; fixed stoichiometry is the value set under Stoichiometry.

Each affinity column chooses its concentration unit independently from its own magnitude. For example, Kd1 may be shown in nM while Kd4 is shown in µM. The column header and its values always use the same unit; this is display scaling only and does not change fitted or saved values.



Project Information

File > Project Information... lists and counts the experiments, Analysis Results, and saved reports in the current project, including inactive experiments. Saved reports hold their selections, presentation settings, study context, and approved interpretation text; the project does not embed exported PDFs or duplicate measurements for each report.

Use the experiment rows to compare temperatures, cell and syringe concentrations, and integration progress. Result rows identify the fitted experiments and their saved temperatures, alongside result validity and binding assessment.

Select a saved report to see its Analysis Results, experiment counts, result availability, whether interpretation text is saved, and any subtitle. **Open** restores its settings and interpretation; **Rename** changes its project entry. **Details** shows the full saved metadata and text, and **Copy All** copies its saved user-entered text. **Delete** removes the report after confirmation. Renaming changes both the saved name and the printed report title. Clear the name to use an automatic name based on its result. **Last saved** records the last save that changed the report; opening, previewing, exporting, and saving the project do not change this date. Older reports may have no recorded save date.

Deleting a report keeps experiments, Analysis Results, and exported PDFs. Reports with missing results can still be inspected, copied, and deleted.

Analysis Report

Tools > Analysis Report... creates a complete, printable report from one or more saved Analysis Results. The same tool is available as **Export Analysis Report...** in result-specific menus, where that result is initially selected. Use **Select report contents...** to choose results and optional supporting experiments together. Opening the tool starts a new report draft. Use **Saved Reports...** in the inspector or **Project Information** to open a saved report and restore its selections, settings, study context, and interpretation. Changing the selected results edits the current draft and retains its interpretation text.

The source experiments of tandems in the selected results may also be selected automatically for a new report. Uncheck any you do not want; saved supporting selections are restored when the report reopens. The picker marks each as **Buffer reference**, **Tandem source**, or both. Choose results and supporting experiments together, then select **Apply**.

Configure the report

The experiment parameter overview focuses on binding model parameters (N or α, K_d, and ΔH, as applicable), relevant conditions, and RMSD. Full thermodynamic values appear in the detailed parameter tables.

Choose a title, an optional subtitle, energy and temperature units, and an **Uncertainties** style. Reports default to **SD + 95% CI**. The selected style controls uncertainty in fitted-parameter tables and applicable analysis plots; the thermodynamic summary chart shows available saved 95% confidence intervals. Combined spread, approximate propagated intervals, and model-estimated uncertainty keep their separate labels: an approximate combined interval does not have established 95% coverage. Presentation choices are stored when you save the report.

Traceability Mode adds a report preparation and sign-off block with the preparer, generation time, and blank preparer-signature and signing-date fields, plus saved **Analysis operator** names, result identifiers, and recorded experiment and sample identifiers. Enter a **Report ID** to identify the report in its signature and report details; the ID is saved with the report and a blank ID appears as **Not recorded**. The report preparer is the current operator when the preview is built; the analysis operator identifies who created or last updated the fit, and can be different. A blank preparer name appears as **Not recorded** while the PDF Author field remains empty. Experiment dates are reported only when they come from the data file or were entered by the user. An experiment that only has a filesystem timestamp shows that date marked **(file system date)**, which does not establish when the experiment was run. If the current operator changed after the preview was built, export automatically rebuilds and refreshes the preview with the current name before writing the PDF. Missing experiment and sample IDs appear as **Not recorded** only in Traceability Mode; otherwise, blank IDs are omitted. Missing IDs do not prevent report creation or export.

Each experiment's parameter table labels values **Fixed**, **Fitted**, or **Derived** based on the saved solution and global constraints. Fixed estimates retain their values and units but do not show uncertainty decoration.

Optional content includes **Injection tables**, **Condense repeated experiments**, **Expanded explanations**, **Parameter correlations**, and completed advanced analyses. Expanded explanations adds contextual notes for approximate summary uncertainty, weighted fit diagnostics, and others.

Condensing repeated experiments replaces repeated acquisition and processing details with a reference to the first occurrence, while retaining each result's figures, fit, and scientific context. The analyses you select are included from their saved results; report creation does not rerun them.

Save a report

Save Report stores the current draft in the project, updating the opened report or creating a new one. **Save Report As...** stores a separate copy and switches to that copy, keeping the original report unchanged. Both include selections, presentation settings, study context, and interpretation. The title supplies the saved name; an automatic or blank title uses a name based on the report contents.

Saving a report marks the project unsaved. Use **File > Save** to write the project to disk. Project save commands, including **Cmd-S** on macOS or **Ctrl-S** elsewhere, save the project without saving edits still in the report draft. Preview and PDF export also use the current draft without saving it. When leaving an edited draft, choose **Save**, **Discard**, or **Cancel**.

Write and preview an interpretation

The builder provides an editor for your remarks. A formatting hint shows the available Markdown shortcuts: `##` section headings, `###` subsection headings, `-` bullets, `**bold**`, and `*italic*` emphasis. Use **Interpretation / Preview** to switch between writing and the rendered report. **Preview** shows the current report, **Update Preview** refreshes it after edits, and **Export PDF...** saves it. Results with warnings or outdated inputs remain exportable with a prominent notice. A structurally unusable result shows a validation error and cannot be previewed or exported.

Editing an interpretation makes it user-written and current, including when it began as generated text. Its previous AI source details and freshness status are replaced. Text edits remain unsaved until **Save Report** or **Save Report As...**, even if you undo them back to the original wording.

If included results change after the interpretation was written or confirmed, the builder asks you to check that the text is still accurate. This includes changes to values, uncertainties, assessments, names, or comments, and changes to which results are included. Choose **Confirm**, or edit the text, then save the report. The notice applies to both user-written and generated text and returns after a later result change. **Confirm** preserves the text and its source; for generated text it does not change the separate **Current / Out of date / Cannot verify** status or printed freshness warning.

Report contents and layout

The report is a printable A4 PDF. It starts with a front page that lists the included results and the contents, followed by a chapter for each result. An approved interpretation follows the front page. Results are referenced as **1**, **2**, and so on, and their experiments as **1A**, **1B**, **2A**, **2B**, and so on. Figures and interpretation use these same references. Results remain separate even when they use the same experiment.

The conditions beneath each experiment heading show nominal initial concentrations in syringe ▸ cell order (for example, `100 µM ▸ 10 µM`) and the target temperature. Recorded buffers with pH, salts, competitor concentration, and explicitly entered ionic strength are also included; Experiment details retains available concentration uncertainties and measured temperature.

The analysis summary identifies the model, options, scope, constraints, fixed settings, fit details, and member estimates. Standard and Diagnostic output follow the binding-assessment policies: no-binding values are suppressed in Standard output, unavailable values use —, and combined values are never recomputed from an eligible subset.

Each experiment's **Fit details** include the Wiseman c-value, $c = NM_0/K_d$, calculated from the initial cell concentration M_0 and the fitted N and K_d (an apparent value from the apparent K_d for Competitive Binding, and one value per site class or sequential step); the curve shape usually determines K_d well only for c roughly between 1 and 1000.

Combined parameters states its **Evaluation temperature**, using the default Parameter Evaluation temperature. This is independent of an edited result-view temperature and the saved fitting reference. Evaluation uses stored relationships without refitting.

The thermodynamic chart includes each eligible result member and active binding step, identifying any omitted no-binding members. Its whiskers show available saved 95% confidence intervals independently of SD availability; SD values remain available in the tables. Fixed parameters have no uncertainty bars. In warning summaries, “N of M experiments” counts affected eligible members against all members of that result; overview fit warnings exclude no-binding members, while Diagnostic appendix warnings retain them. Experiment sections include the thermogram where available, a final figure, any fit warnings for that experiment, fitted and derived parameters, and selected correlations. **Fit details** reports dimensionless c-values where the model defines them. For one- and two-site models, the Wiseman value is $c = N[\mathrm{cell}]_0/K_d$; syringe correction uses the fixed site count. Sequential fits report $[\mathrm{cell}]_0/K_d$ for each active step, while competitive fits use the apparent $K_{d,\mathrm{app}}$. When **Injection tables** is selected, the report includes each injection's inclusion state, volume, concentrations, heat, measurement SD, fitted heat, and residual. Fit and residual columns use the reported member’s saved solution; residual means observed heat minus predicted heat. Supporting-experiment tables show observations only, even if the experiment has an attached fit. An unavailable thermogram is identified in the report.

Each experiment's processing or data-availability details include the injection bookkeeping saved with its fit. When experiments in one result use different bookkeeping methods, the result overview and the report warnings flag it. The analysis summary lists the buffer subtraction saved with each fit; experiment details show the experiment's current buffer subtraction among its attributes. Parameter tables identify fixed estimates and omit uncertainty intervals for them. Processing details, observations, and injection tables describe **Current experiment data**, while fitted parameter values come from the saved result. Supporting experiments have no saved fit settings, so their subtraction details are omitted; correction provenance remains available where relevant.

Competitor properties shows source values captured for the reported fit when available. Selecting a source alone does not capture its properties; capture occurs when a fit uses **From attributes** for competitor affinity or enthalpy.

Baseline method identifies the current spline interpolation and actual point count, or the polynomial degree or segmented baseline type. Spline density describes the current points, including manual edits and conversions.

When an experiment has no raw thermogram, **Data availability** identifies its input as imported integrated heats. Baseline and integration details appear only for experiments with a thermogram.

Selected experiments not already represented by a result appear once as **Supporting experiments**, labeled **S1**, **S2**, and so on. They contribute existing traces, processing, heats, metadata, dates with their source, attributes, comments, and optional injection tables; their heats

are explicitly shown without a fit. Represented experiments cannot be selected again as supporting data. Automatically selected buffer references remain optional, and the report does not judge whether they are appropriate blanks. Creation does not fit, reintegrate, or perform subtraction.

Source file identifies each experiment's source filename and format, or marks the format unknown.

A single appendix at the end of the report lists every experiment once, with the labels it is reported under and any role as a buffer reference or tandem source (for example **Tandem source for 1A, 2B**), followed by report warnings and software details.

Source links are recorded when the Experiment Merger creates a tandem. If any of those sources are no longer in the project, a **Tandem sources** item in the tandem's experiment details gives the count. Older tandems and files that already contain concatenated runs have no source links; add their source experiments manually if needed.

Generate an automated interpretation

Choose **Generate interpretation...**, enter a **Main question** and **Additional context**, and select **Provider** and **Interpretation depth**. Review the selected results and supporting experiments. **Evidence level** controls fitted result detail: **Results**, **Injection data**, **Processing details**, or **Thermograms**. Each level includes the lower levels. Injection data is the default where permitted; Public access uses Results. Standard access permits Injection data, and Advanced or Administrator access also permits Processing details and Thermograms. Thermograms requires a fresh selection each time the dialog opens. Supporting experiments always provide summary information; they do not send injection rows or traces. The inspector identifies user-written or generated text and its freshness.

Review the returned draft, then choose **Use in report** to add it to the report. Edit it in the report builder if needed. Check its scientific claims and references against the underlying results.

Online generation privacy: Generate sends selected names, comments, conditions, fits, injection evidence, and your question/context to the FT-ITC service, which uses your selected OpenAI or Mistral provider. Thermograms require opt-in. Usage metadata has no automatic expiry; full evidence and generated text are excluded from that metadata. Provider and infrastructure retention are separate, with no guaranteed deletion deadline. Canceling does not retract sent data. Read Data flow and retention before sending sensitive material.

Evidence and interpretation limits

Generation assesses all selected results and supporting experiments together, using the same references as the PDF. Results remain independent, including alternative fits of the same experiment. Supporting experiments contribute summary metadata without being assigned a fitted result or assumed control role. Processing details and Thermograms add processing settings and baseline summaries; injection rows or traces are not sent for supporting experiments. Generation uses existing processing, integration, residual, and uncertainty evidence where available; it does not rerun integration, baseline fitting, or model fitting. When compressed thermograms are included for fitted result members, they contain minimum and maximum power values for each 15-second interval, plus an aligned fitted baseline when available. Exact peak timing, the order of values within each interval, and the detailed waveform are unavailable. The dialog shows the

compact package size against your current request limit and blocks generation when the estimated request is too large. An opted-in thermogram is retained in the request; a size or model-context error is shown instead of silently omitting it.

Generation requests include each tandem's source links; a source experiment's own data are sent only when it is selected for the report.

When generation succeeds, the desktop app sends an **Interpretation ready** system notification. Whether it appears depends on operating-system notification support and permissions. The draft remains available in the generation dialog for review.

Completed structuring, electrostatics, and protonation analyses are included when available for the selected results. Results includes fitted values, uncertainty, assessments and fit/residual summaries. Injection data adds acquisition details and injection-by-injection heat and fit comparisons. Processing details adds integration boundaries, baseline statistics and controls, and historical fit-input snapshots where available. Thermograms adds compressed signal and aligned baseline traces.

If the selection contains outdated or failed analyses, a warning gives you an opportunity to review them before continuing. The report distinguishes saved fit inputs and estimates from current observations; some comparisons are unavailable when the saved fit cannot be matched to the current data. An interpretation can identify potential concerns, but it does not certify the experiment or establish a mechanism. The service may suggest references from details such as a paper title, journal, or year, but it does not search the web.

Markdown tables require a header and separator row. Use a few columns to keep them readable; unsupported formatting may appear as literal text in the report.

The generated draft is read-only in the generation dialog. Choose **Use in report**, then edit it in the report builder if needed. Canceling, closing the dialog, or encountering a service error leaves previously approved text unchanged. Study context, evidence settings, approved text, and generation details are saved with the report. Changes to selected results, supporting evidence, context, or settings can mark an untouched generated interpretation as out of date. In Traceability Mode, user-written interpretations show their user source and writing time; untouched generated interpretations show their generation details.

Save a local copy of interpretation evidence

Use **Save package** in the generation dialog to save a local ZIP of the evidence, question, context, and settings selected for that request. Saving the ZIP does not contact the interpretation service or run new fits. The archive contains the full local evidence, including supported injection data and compressed thermograms, together with the exact compact payload prepared at the selected evidence level. The level limits what is sent to the service; the local archive can contain more data. Review it before sharing.

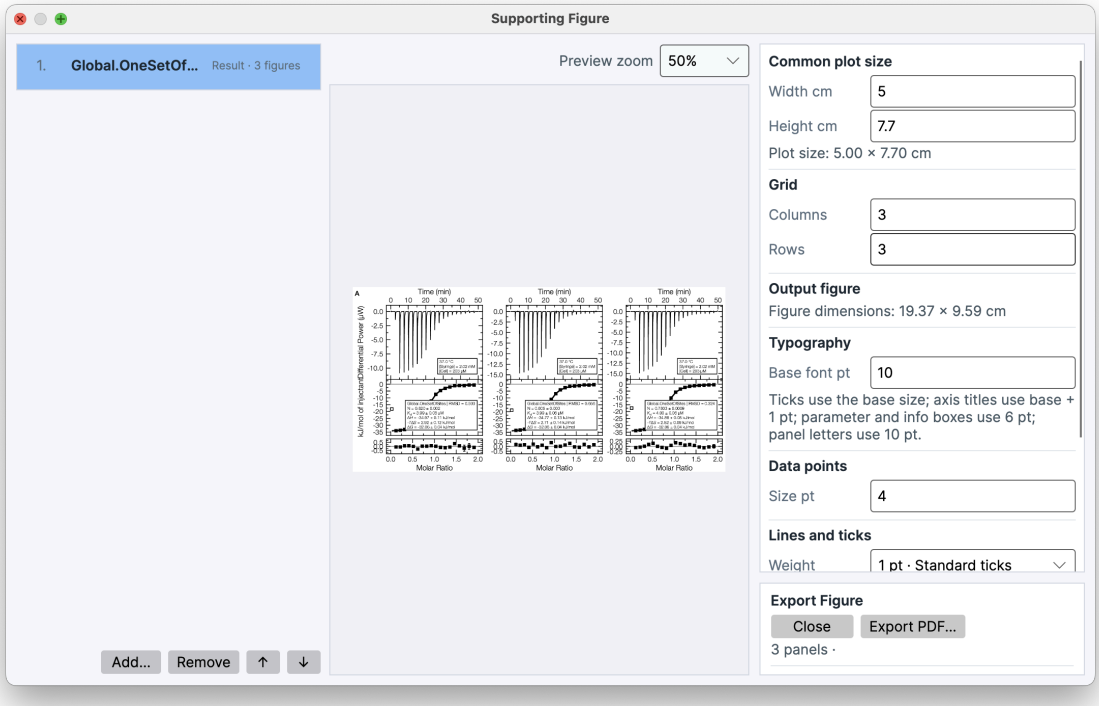
Troubleshoot automated interpretation

If generation fails, note the visible error and use **Copy Support Report** when contacting support. Local report editing and PDF export remain available if online generation is unavailable.

Supporting Figure

Supporting Figure... composes figures from Experiment Data and Analysis Results into a multi-panel canvas. The **Figure order** list defines source order; **Add...**, **Remove**, **Up**, and **Down** change the composition. Closing and reopening the tool retains the selected sources, their order, plot size, grid, typography, display settings, and preview zoom for the current document during this app session. The source picker can filter available experiment and result figures by name or type.

Set the plot size, grid columns and rows, font size, point size, line weight, and tick style for the composed figure. Extra panels continue on additional PDF pages. **Light** and **Standard** provide different line and tick weights. **Panel letters** labels panels A, B, and so on; **Panel titles** can add experiment names. **Group result figures** and the information-box controls adjust how results are presented. Review the preview, then choose **Export PDF...**.



Printing

File > Print prints the active graph or figure through the operating system's print workflow. The active target can be the overview thermogram, processing graph, analysis graph, result graph—including an available **Correlation** matrix—or Final Figure, depending on the selected workspace. The operating-system print dialog supplies the available printer and PDF destinations; the graph content comes from the active application view.

CHAPTER 10

Tools

Design simulated titrations, subtract buffer controls, and merge standard or back-mixed tandem experiments.

Last verified: 2026-09-23

Use **Experiment Designer...** to explore a simulated experiment before collecting data, **Buffer Subtraction...** to account for heat seen in a reference experiment, and **Experiment Merger...** to join consecutive parts of a titration. The sections below explain what each tool takes as input and what it changes in your project.

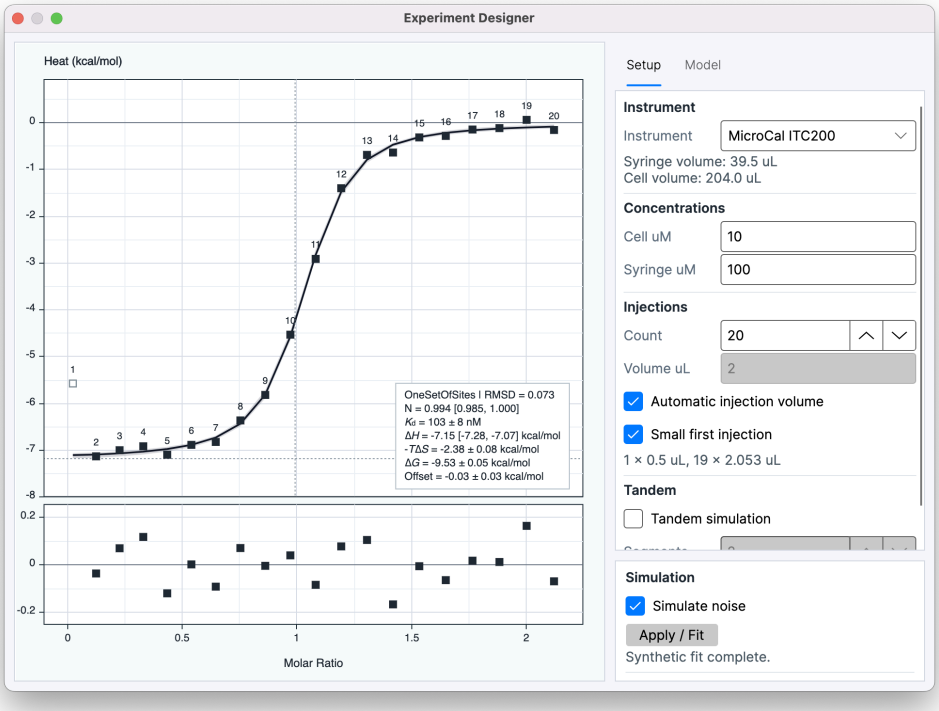
Open these tools from **Tools**. Designer simulations stay in their window; Buffer Subtraction stores corrections on targets; Experiment Merger creates a new processed Experiment Data item. See Workspace for selecting project items.

Experiment Designer

Experiment Designer... creates a synthetic titration from instrument, concentration, injection, and model parameters. The graph updates as the design changes. The designer does not add the synthetic experiment or its fit as a project item.

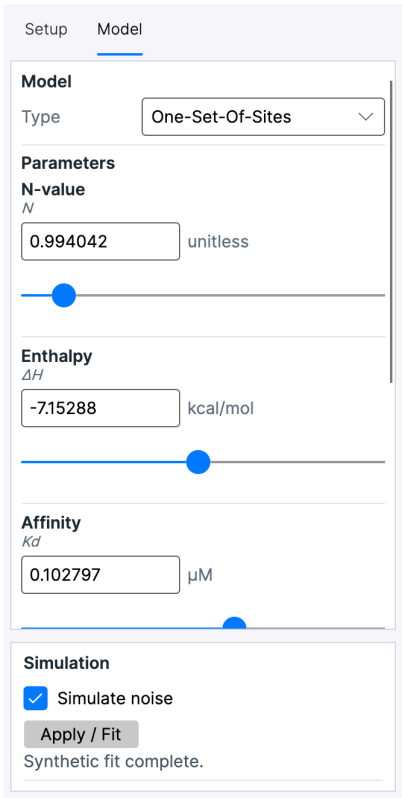
Setup and model controls

The **Setup** tab contains **Instrument**, **Cell uM**, **Syringe uM**, injection **Count**, and **Volume uL**. Count and volume define the injection schedule. **Automatic injection volume** derives the injection volume from the selected instrument's standard syringe volume and injection count. **Small first injection** gives the first injection of each load a smaller volume and marks it excluded. The **Simulation** section contains noise controls. **Tandem simulation** creates consecutive loads, with **Segments** setting the number of loads. The designer uses its tandem back-mixing model between loads.



The **Model** tab contains **Type**, exposed model **Parameters**, and model-specific **Options**. Parameter sliders use practical editing ranges; enter a value in the text field to set a value beyond a slider's range.

For Competitive Binding, enter the total competitor concentration and its dissociation constant (**Kd**) in the displayed concentration unit. Enter competitor binding enthalpy in the displayed energy unit per mole. For example, K_d = 1 µM and enthalpy = –40 kJ/mol are entered as **1** and **–40** when those units are shown.



Buffer Subtraction

BEFORE YOU BEGIN:

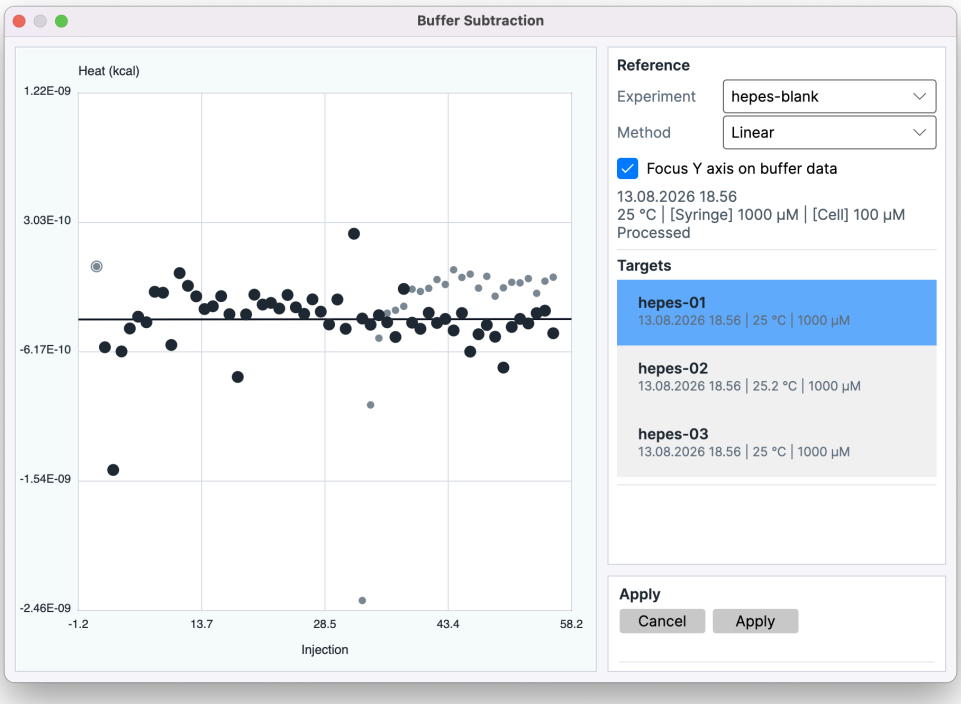
Buffer subtraction operates on integrated injection heats. Use experiments with available heats to inspect the correction in the preview; thermogram inputs need processing to supply those heats.

Buffer Subtraction... models background heat from one reference experiment and applies the resulting correction to one or more target experiments. The reference selector shows experiment metadata and a **Processed** or **Not yet processed** status. The target list excludes the selected reference and tandem experiments, and supports multiple targets. Buffer subtraction is currently unavailable for tandem experiments, including through Experiment Details. The correction uses the reference's available integrated heats; if the reference is processed later, target corrections update. Processing is described in Processing.

The **Method** selector contains **Matched**, **Linear**, and **Exp. decay**:

- **Matched** evaluates the reference heat at each target injection number, using nearby included reference injections when the matching injection is unavailable.
- **Linear** fits a line through valid, included reference injections and evaluates it across target injections.
- **Exp. decay** fits an exponential-decay model through valid, included reference injections when enough points are available.

The preview graph shows reference and target heats and the selected subtraction model. Reference-point inclusion changes the points available to the fitted methods. **Focus Y axis on buffer data** changes only the preview range. A continuous model line appears for **Linear** and **Exp. decay**; **Matched** is represented by injection-level reference values.



Apply stores the reference and method on each target. The corrected heats are then used for subsequent fitting and export while the original integrated heats remain unchanged. The reference Experiment Data becomes inactive. Changes in its processing or injection inclusion update the target corrections. The subtraction is project data and can affect the validity of dependent results.

When a target is saved without its reference experiment, for example with **Save Selected...**, the reference's included integrated heats are stored with the target. The target's attribute then shows **Missing reference (stored values retained)**. The correction stays in effect when the target is reintegrated, but the stored reference heats cannot be reprocessed. Opening the reference experiment again, or applying another reference, replaces the stored values.

Corrected heat and uncertainty

CALCULATION:

$$q_{i,\mathrm{corr}} = q_{i,\mathrm{target}} - q_{i,\mathrm{ref}}$$
$$\sigma_{i,\mathrm{corr}} = \sqrt{\sigma_{i,\mathrm{target}}^2 + \sigma_{i,\mathrm{ref}}^2}$$

The selected method determines the reference value evaluated for injection i. $q_{i,\mathrm{target}}$ is the target heat, $q_{i,\mathrm{ref}}$ is the corresponding reference value, and $q_{i,\mathrm{corr}}$ is the corrected heat used for fitting and export. The σ values are standard deviations (SDs). The corrected heat's SD combines the target and reference SDs as independent errors; subtracting the heats does not subtract their SDs.

With **Matched**, the reference SD comes from the matched injection. If the method uses the average of the nearest included injections on either side, its SD is half the square root of the sum of their squared SDs. With **Linear** or **Exp. decay**, the subtracted reference value is the fitted line or curve, so the reference SD is the uncertainty of that fitted value at injection i, not the scatter of single reference heats. It is calculated from the scatter of included reference heats around the fit (not their individual error bars) and from how well the included injections determine the fit at injection i. Because the fit averages over many injections, this SD is usually much smaller than the scatter and is largest near the ends of the reference range. The same fitted curve corrects every injection, so this part of the error is shared between injections rather than independent. If the fit has no degrees of freedom left to estimate that scatter, the model assigns a reference SD of zero; this does not establish that the correction is exact.

Experiment Merger

BEFORE YOU BEGIN:

Select at least two source experiments with thermograms. The merger uses a source's baseline-corrected trace when available and its raw trace otherwise. It processes the newly merged Experiment Data automatically.

Source experiments must have no buffer subtraction. Remove any buffer-subtraction attribute in **Experiment Details** before merging or scanning. This restriction applies to every merger mode.

Experiment Merger... joins two or more eligible thermogram experiments from consecutive segments of a tandem titration. The source list contains thermograms that are not already tandem experiments. Selection order defines segment order; **Up** and **Down** reorder selected rows.

The merge **Mode** selector contains:

- **Simple tandem**, which concatenates the segments using the standard concentration progression without a user-selected back-mixing correction.

- **Fixed back-mixing**, which applies one configured mixing fraction at every segment transition. With three or four selected experiments, you can instead set an individual fraction for each reload.
- **Auto back-mixing**, which estimates the reload mixing fractions from the selected experiments and is available for up to five source experiments.

Back-mixing controls include **Dead vol. uL**, the **Mixing** fraction, and **Remove titrated overflow**. Dead volume represents the filling-stem or overflow volume above the active cell volume. The overflow control records whether titrated overflow was removed between segments. In Fixed mode, the shared slider supplies the fraction. With three or four experiments selected, enable the individual-fractions option to set **Reload 1**, **Reload 2**, and, for four experiments, **Reload 3** separately. Auto mode estimates the values from the selected data; these are inferred corrections rather than direct measurements of mixing.

Choose **One-site** when a one-set-of-sites model is appropriate, or **Model-free** to match the two sides of each reload without a binding model. Model-free needs at least three included injections among the six nearest on each side. Compare criteria for noisy data or reloads near saturation.

Create produces a new processed Experiment Data item. Its thermogram samples are time-shifted and concatenated, its injection sequence retains segment boundaries, and its segment metadata stores the calculated starting active-cell and active-titrant concentrations. The merged item's comments retain the first source's notes and append the selected tandem mode, the auto criterion when used, the back-mixing settings, and the selected fractions in reload order. Source experiments remain separate and are not changed by creation; the new item is marked as a tandem experiment and is not eligible as a later merger source. It is a snapshot and does not update if its source experiments are subsequently edited. The resulting item can be fitted through Analyze Data.

When the first source experiment has an external experiment ID, the merged experiment's ID is that value followed by `-concat`; if its ID is blank, the merged ID stays blank. The cell and syringe sample/batch IDs are copied only when every source has the same nonblank value; otherwise the merged IDs are blank. The merger does not open the import review, so select the merged experiment and open **Details > Identifiers** to add or change IDs.

The new experiment uses the **Injection bookkeeping** method selected in **Preferences > Processing** for dilution during each injection. That choice is separate from the merger's between-segment back-mixing setting. See Injection bookkeeping for the methods and their limits.

What happens at a reload

In **Fixed back-mixing** and **Auto back-mixing**, the model tracks the active cell and the dead/overflow volume separately. It starts with the same macromolecule concentration in both and no syringe ligand in either. During a segment, injections change the active-cell concentrations and displaced material enters the dead volume. At the transition to the next segment, the model first removes overflow if **Remove titrated overflow** is selected, then mixes the chosen fraction of the remaining dead volume with the active cell. The resulting active-cell concentrations become the starting concentrations for the next segment. This changes the calculated concentrations used for fitting; it does not add a heat peak or alter the measured thermogram.

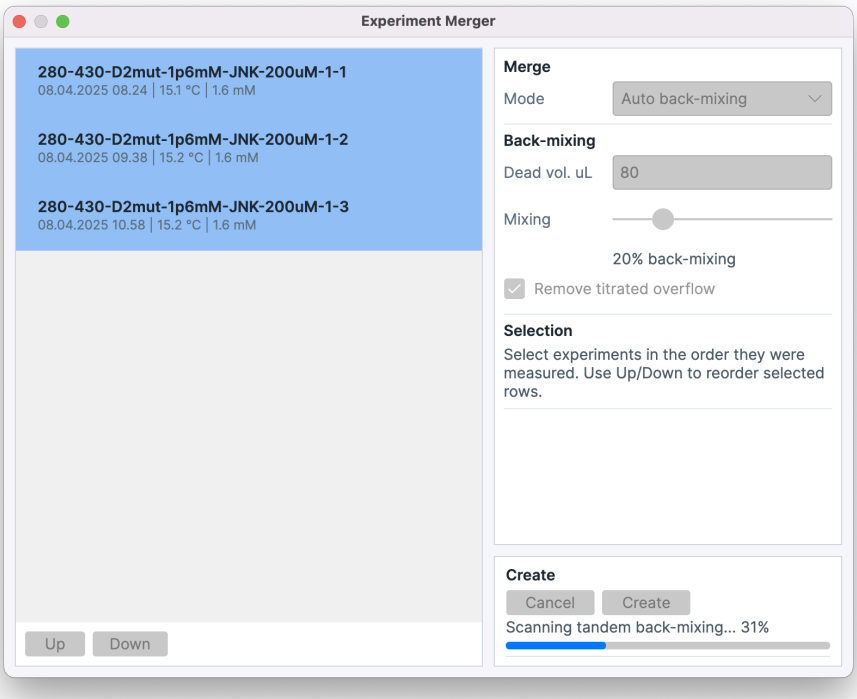
CALCULATION:

For each species (macromolecule and syringe ligand), let V_{cell} and C_{cell} be the active-cell volume and concentration just before mixing, V_{dead} and C_{dead} the remaining dead-compartment volume and concentration, and f the selected mixing fraction.

$$V_{\text{mix}} = f \cdot V_{\text{dead}}$$

$$C_{\text{next}} = \frac{V_{\text{cell}} \cdot C_{\text{cell}} + V_{\text{mix}} \cdot C_{\text{dead}}}{V_{\text{cell}} + V_{\text{mix}}}$$

If overflow is removed, the model first removes a volume equal to the injections delivered in the completed segment, capped at the available dead volume. It assumes that compartment is well mixed, so removal takes the same proportion of each species. The formula then uses the volume and concentrations left behind. A mixing fraction of 0 leaves the active-cell concentrations unchanged; a fraction of 1 includes all remaining dead-compartment liquid in the mixing calculation. The active-cell volume itself stays fixed, and the model retains the remaining material in the dead compartment for later transitions.



INTERPRETATION:

A configured or automatically fitted back-mixing fraction is a model-based correction, not a direct measurement of liquid mixing that occurred between runs.

Automatic back-mixing calculation

One-site minimizes the combined titration's one-set-of-sites RMSD. Model-free solves reloads in order, keeping earlier fractions. Each candidate recalculates concentrations and fits one unweighted, monotonic cubic to heat per mole of injectant against midpoint molar ratio, using included injections among the six nearest positions on each side. Excluded injections still contribute their delivered volume to bookkeeping.

The cubic must keep rising or falling across the window, so it assumes the local isotherm does not turn. Where fractions are poorly distinguished, a weak, chosen preference favors 5%; this is not measured mixing information. The small local window makes model-free estimates less precise with noise or near saturation.

CHAPTER 11

Settings and defaults

Configure shared defaults for display, processing, fitting, and export.

Last verified: 2026-09-23

Preferences set the starting behavior for future work, such as display units, processing choices, and export formatting. Changing a default does not by itself redo an existing fit or processing decision saved in a project.

Open **Preferences...**, edit the relevant tab, and choose **Apply**. Defaults initialize new or reset work; display settings change presentation and export settings change output. Saved project settings remain distinct. See Installation, files, and projects for saving and recovery.

Restore Defaults stages the built-in values in the window. **Apply** saves the staged values as application preferences. **Cancel** closes the window without saving staged edits.

Apply saves only the settings you edited and updates their related views. Untouched settings retain changes made elsewhere while Preferences was open. **Restore Defaults** explicitly stages every built-in preference, including settings not shown in that application's window.

General

SETTING	EFFECT
Enable Traceability Mode	Requires an operator name. While enabled, FT-ITC asks you to confirm the operator once at each launch; see Traceability Mode at startup. Off by default.
Operator name	Name captured on newly created or successfully updated Analysis Results and when a report preview is generated. Existing results and a current preview keep their saved attribution. Required while Traceability Mode is enabled.
Prompt for experiment and sample IDs after import	While Traceability Mode is enabled, asks for experiment, cell sample, and syringe sample IDs after data files are imported. All IDs are optional. The setting is kept, but has no effect, while the mode is off.
Energy units	Select **Joule** or **Calories**. Displayed values automatically use the base or kilo prefix according to the magnitude of the displayed central values; empty groups default to kJ or kcal.

SETTING	EFFECT
Concentration unit	Shared default unit for concentration entry, parameter display, and supported result-table concentration fields; format-specific data exports use their documented units.
Temperature units	Sets the Celsius or Kelvin default for Results and Analysis Result Exporter tables. Each keeps its own choice for the current application session. Changing this preference resets both choices immediately.
Designer instrument	Shared default instrument for the Experiment Designer and its instrument-specific volumes.
Number precision	Controls numeric presentation: **Strict**, **Standard**, **Single decimal**, or **All decimals**. Under every setting, an estimate, SD, or confidence-interval bound with a magnitude of 10^{10} or more is shown in scientific notation with up to six significant digits, and an infinite bound is shown as ∞ or −∞. This changes presentation only. It does not set export decimal places.
Uncertainty display	Shared uncertainty presentation: **Automatic**, **Standard deviation**, **Confidence interval**, **SD + confidence interval**, or **None**. Automatic uses the confidence interval for a quantity whose stored 95% interval is materially asymmetric around its best-fit value; otherwise it uses SD. This changes presentation, not the underlying fit or calculated uncertainty.
Reference temperature (°C)	Temperature used where a result or derived quantity is evaluated at a reference temperature. It is also the proposed experiment temperature when importing `.dat` and `.aff` files, which do not record one.
Minimum temperature span (°C)	Minimum temperature variation required for temperature-dependent result analyses.
Minimum ionic-strength span (mM)	Salt analysis requires three ionic strengths, selected in increasing order with each more than half this span above the previous selected value.
Include buffer in ionic-strength calculation	Includes buffer contribution when ionic strength is calculated.
Check for updates and online resources on launch	Controls launch-time online checks; it does not disable local analysis.
Confirm remove/delete actions	Controls confirmation before remove or delete commands.
Enable autosave	Enables periodic recovery copies.
Interval (minutes)	Sets the autosave interval. The built-in default is **5 minutes**.
Maximum files	Sets the number of autosave files retained. The built-in default is **10 files**.

SETTING	EFFECT
Prompt to recover after an interrupted session	Controls whether the application presents an available autosave recovery after an interrupted session. Recovery prompting is enabled by default.
Open Autosave Folder	Opens the folder containing autosave recovery files.

Display units do not change interchange-format units. Automatic energy scaling uses kJ or kcal for empty, zero, non-finite, or at-least-100-unit groups. Fixed figure and result-export overrides are selected in their dialogs; see Figures and export.

Processing

SETTING	EFFECT
Injection bookkeeping	Selects **Discrete displacement** (default and recommended starting point for ordinary pulse injections), **Ideal continuous mixing** (for slow injections or opposite-limit comparisons), or **MicroCal** (to reproduce or compare MicroCal analyses) for new data. Change an existing ordinary experiment explicitly in Experiment Details; rebuild a tandem experiment to change its method. Saved fits are not automatically migrated.
Buffer subtraction	Sets the default buffer-subtraction model: **Matched**, **Linear**, or **Exp. decay**.
Discard integration regions for baseline	Controls whether existing integration regions are excluded from baseline construction.
Recalculate concentrations on integrated-heat import	Sets the initial **Recalculate concentrations and ratios** choice for `.dat` and `.aff` imports. You can change it for an import without changing this preference. Recalculation uses the imported injection volumes and experiment concentrations; it does not create a thermogram or repeat baseline correction and peak integration. `.dh` concentrations are always calculated.
Point density	Sets the default spline point density: **Sparse**, **Balanced**, or **Dense**.
Handle mode	Sets the default spline handle calculation: **Mean** or **Median** in the preferences window.
Allow spline point time dragging by default	Controls whether spline points can be moved in time by default.
Copy integration start with selected region	Controls whether copying an integration region also copies its start time.

For baseline, integration, and locking choices, see Processing. For method assumptions, see Injection bookkeeping. Stored experiment processing is not replaced by changing defaults.

Fitting

SETTING	EFFECT
Default solver	Sets the starting optimizer: **Nelder-Mead [SIMPLEX]** or **Levenberg-Marquardt**.
Error estimation	Sets the default uncertainty method: **None**, **Bootstrap residuals**, **Leave-one-out**, or **Profile likelihood**.
Bootstrap iterations	Sets the number of residual-bootstrap refits. Leave-one-out uses one refit per deletion and profile likelihood does not use this count.
Optimizer tolerance	Sets the solver tolerance preset: **Fast**, **Relaxed**, **Balanced**, **Strict**, or **Very Strict**.
Max iterations	Sets the maximum number of optimizer iterations. The built-in default is **20,000**, including after **Restore Defaults** is applied.
Parameter limits	Sets the default parameter-limit policy: **Standard**, **Extended**, or **No limit**.
Use injection-error weighted fitting	Controls weighting of injection observations by their estimated errors.
Include concentration uncertainty in bootstrap	Includes concentration uncertainty in residual-bootstrap resampling. Leave-one-out and profile likelihood keep concentrations fixed.
Create single-experiment analysis result	Controls creation of an Analysis Result after a usable single-experiment fit.
Create global analysis result	Controls creation of a combined Analysis Result after a usable multiple-experiment fit.
Auto-open new analysis result	Controls whether a newly created result is opened automatically.

Concentration uncertainty and parameter unlocking apply only to residual bootstrap. See Single-experiment fitting for model and uncertainty interpretation.

Saving a project or applying Preferences preserves the Analysis inspector's solver, uncertainty method, bootstrap count, weighting, and concentration-uncertainty choices. Defaults for these choices are loaded at application startup or when you choose **Restore defaults** in Analysis.

Export

SETTING	EFFECT
Selection	Sets the default export scope: **Selected experiment**, **Active experiments**, or **All experiments**.
Decimals	Sets the number of decimal places in exported numeric tables.
Export baseline-corrected data	Includes baseline-corrected data in data exports.
Export fit points with peaks	Includes fitted peak points with exported peak data.
Molar ratio	Includes molar-ratio values in exported tables.
Injection info	Includes injection volume, delay, peak error, integration length, and temperature columns.
Concentrations	Includes cell and syringe concentration columns.
Included state	Includes the injection-inclusion state.
Peak heats	Includes integrated peak heats.
Fit values	Includes fitted values.
Width cm and **Height cm**	Set the default final-figure dimensions. The built-in dimensions are **6.5 × 10 cm**.
Publication font	Selects the publication figure font where the platform provides this selector.
Show residual graph	Includes the residual graph in final figures.
Show residual graph gap	Includes the gap separating residual and fit panels.
Unify residual graph axis	Uses a common residual-axis scale across applicable panels.
Fit line	Sets fit-line smoothing: **Smooth**, **Spline**, or **Linear**.
Show parameter box by default	Includes the parameter information box in new final figures.
Show experiment details by default	Includes experiment metadata in the figure information box.
Show model info by default	Includes model information in the figure information box.
Auto axes ignore excluded/bad points	Excludes points marked excluded or bad when automatic axes are calculated.
Thermodynamic parameters	Includes thermodynamic parameters in final-figure information.

SETTING	EFFECT
Offset parameter	Includes the fit offset in final-figure information.
Derived parameters	Includes derived parameters in final-figure information.
Temperature	Includes temperature in final-figure information.
Concentrations	Includes concentrations in final-figure information.
Injection delay	Includes injection delay in final-figure information.
Instrument	Includes instrument information in final-figure information.
Attributes	Includes experiment attributes in final-figure information and limits their display to **Used in analysis**, **All**, or **None**.

Export preferences affect newly generated exports and figure defaults; they do not rewrite an existing export or a stored figure configuration. See Figures and export for output formats and units.

The energy-unit preference also sets the initial molar-energy unit in the Integrated Peaks, Combined Data, and ITCsim export dialogs: kJ/mol for Joules or kcal/mol for Calories. You can choose another unit for each export without changing the preference.

> **PLATFORM NOTE:**
> The **Publication font** selector is available on Windows and Linux, with choices of **Native**, **Inter**, and **Liberation Sans**. Some macOS releases use the native publication renderer and do not show this selector.

Interpretation

To request an account with additional interpretation options, use the FT-ITC interpretation service registration page. If eligible, submit your name and email, accept the Terms and acknowledge the Privacy Notice, then verify your email; the access code is sent separately by email. Keep the code confidential. If access-code delivery fails, resubmit the form with the same email address to retry delivery of the same code; a resend cooldown applies. Contact `support@ft-itc.org` if the code is exposed or still does not arrive.

Without a verified account code, automated interpretation uses Public access and the **Fast** interpretation preset.

Enter the account code and choose **Verify Access** to load the options available to your account. When additional presets are available, choose one from **Interpretation depth**. Preferences show available account details and remaining quota. The code stays masked and is checked when an interpretation is generated.

CHAPTER 12

Glossary and equations

Terminology, keyboard conventions, calculation symbols, equations, and references.

Last verified: 2026-09-23

Use the tables to look up terms and symbols, and the equation index to reach each calculation beside its workflow and assumptions. Start with Quick start for instructions.

Workspace terms

TERM	MEANING IN FT-ITC ANALYSIS
Experiment Data	One imported or application-created ITC dataset, including its details and analysis state.
Data / Results	The workspace list containing experiments and stored Analysis Results.
Selected	The item currently shown or targeted by a selection-specific command.
Active experiments	Experiments included in a multi-experiment operation.
Solution	A fitted model state for one experiment.
Analysis Result	A stored fit result with one or more member solutions and the fit state used to create it.
Attribute	Experiment metadata used for organization or analysis context.
Valid result	An Analysis Result whose recorded fit inputs still match the current project state.
Correlation view	Pearson correlations between fitted parameter coordinates across residual-bootstrap refits that have finite values for every displayed parameter, with diagnostics for simulation precision.
Pearson correlation	A value from –1 to +1 describing linear co-variation between two fitted coordinates.
Monte Carlo precision interval	Approximate Fisher-transformation interval describing how precisely the correlation coefficient r is estimated from the finite retained-refit ensemble; it is not uncertainty for either fitted parameter or a model-validity assessment.

TERM	MEANING IN FT-ITC ANALYSIS
AIC / AICc	Fit-comparison scores that account for parameter count; smaller is preferred only for comparable observations and likelihood conventions. They do not establish model adequacy. See Information criteria for weighting, variance estimation, member scope, and AICc's small-sample approximation.

Uncertainty terms

TERM	MEANING IN FT-ITC ANALYSIS
Best-fit value	Parameter value from the primary optimization. Resampling estimates uncertainty around this value but does not replace it with a resampling mean or median.
Standard deviation (SD)	For residual-bootstrap parameters, the root-mean-square deviation of retained refit values from the primary best-fit value, transformed or propagated for display. Profile-likelihood results use the equivalent display scale defined below.
95% confidence interval (CI)	For a residual-bootstrap parameter, the 2.5th and 97.5th percentiles of the retained refit distribution, transformed for display. Profile-likelihood intervals use likelihood-threshold endpoints. Derived quantities use the application's uncertainty-propagation rules. Intervals can be asymmetric around the best-fit value.
Automatic uncertainty display	Shows CI for a materially asymmetric stored interval and SD otherwise. It selects a presentation separately for each reported quantity.
Displayed parameter uncertainty	Fitted-parameter uncertainty after conversion or propagation into the quantity shown to the user. The Automatic display decision is applied to this displayed quantity.
Profile likelihood	A 95% interval search that holds one parameter at trial values while refitting the others. A bound reached before the threshold is censoring, not a confidence endpoint. See Profile-likelihood uncertainty for calibration and relative-weight assumptions.
Profile equivalent SD	Symmetric `value ± SD` display scale computed from a complete asymmetric profile interval; it is a display equivalent, not a Gaussian sampling SD.

Keyboard conventions

The application command modifier is **Command** on macOS and **Ctrl** on Windows or Linux.

SHORTCUT	FUNCTION
Command/Ctrl + O	Open data or a project.

SHORTCUT	FUNCTION
Command/Ctrl + S	Save the current project.
Command/Ctrl + Shift + S	Save the project under another name.
Command/Ctrl + E	Open data export.
Command/Ctrl + P	Print the active graph or figure.
Command/Ctrl + Z	Restore the most recently removed item.
Command/Ctrl + I	Invert the Active Experiment Data set.
Command/Ctrl + comma	Open Preferences.
Command/Ctrl + Q	Quit FT-ITC Analysis.
F1	Open **Help and Guide** on Windows and Linux. On native macOS, choose **Help > FT-ITC Analysis Help**.
Enter	Open **Details** for the selected Data/Results item on Windows and Linux.
Left / Right	Change the selected injection in **Process Data**.
Space	Copy the selected integration end-point offset to the next injection in **Process Data**; the start is also copied when that processing option is enabled.

Thermodynamic symbols

SYMBOL	MEANING
N	Stoichiometry or site-number term under the selected model.
α	Syringe-side active-concentration correction when syringe correction is enabled.
K_{a}	Association constant.
K_{d}	Dissociation constant.
K_i	Macroscopic association constant for sequential transition i.
β_i	Cumulative sequential association product $\prod_{j=1}^{i} K_j$.
F_i	Fraction of macromolecule in sequential state MX_i.
$\bar{\nu}$	Mean ligand occupancy in the sequential model.
ΔH	Binding enthalpy change.
ΔG	Gibbs free-energy change.

SYMBOL	MEANING
$-T\Delta S$	Entropic contribution reported by the application at the evaluation temperature.
ΔC_p	Heat-capacity change in a supported temperature-dependent analysis.
q	Integrated injection heat.
σ	Standard uncertainty or spread, according to context.
Offset	Energy-per-mole-of-injectant correction included in the modeled injection heat.

Calculation symbols

SYMBOL	MEANING
$P(t)$	Differential power at time t in a thermogram.
$b(t)$	Estimated baseline at time t.
n_i	Amount injected in injection i.
r_i	Residual for injection i: observed heat minus model heat.
r_{jk}	Pearson correlation between fitted parameter coordinates j and k.
θ_{bj}	Value of fitted coordinate j in complete bootstrap refit b.
σ_i	Processing-derived uncertainty for injection i.
R	Gas constant.
T	Absolute temperature, in kelvin, for thermodynamic relationships.
I	Ionic strength used in salt analysis.
c_{salt}	Recorded concentration of the single salt used for Salt Dependence.
Δn_{H^+}	Fitted slope of observed binding enthalpy against buffer ionization enthalpy: net proton uptake (positive) or release (negative) per binding event.
ΔH_0	Buffer-independent binding enthalpy at the experimental pH: the Protonation intercept, retaining binding-partner protonation heat.
s	Salt-dependence slope: change in ln(Kd / M) per change in ln(Salt concentration / M).
RSS	Raw residual sum of squares, $\sum r_i^2$, over included injections.
Q	Standardized residual sum of squares, $\sum (r_i/\sigma_i)^2$. Weighted AIC/AICc estimates a common variance multiplier as Q/n.
L	Gaussian likelihood used for AIC/AICc.
n	Number of included observations in the information-criterion calculation.

SYMBOL	MEANING
p	Number of free fitted parameters in the saved global model; shared coordinates count once and member coordinates count per member.
K	AIC/AICc likelihood parameter count: $p + 1$ in both weighting modes, including one estimated common residual variance or one estimated multiplier of the injection-error variances.

Equation index

RELATIONSHIP	LOCAL EXPLANATION
Integration and molar normalization	Integration calculation
Processing-derived injection heat SD	Noise calculation and limits
Dilution-driven dissociation of an associated pair	Dissociation
Sequential binding polynomial and ligand balance	Sequential Binding Sites
Thermodynamic conversion and **−TΔS**	Thermodynamic relationships
Fit residuals, RMSD, and weighting	Weight by injection error
AIC, AICc, Gaussian likelihood, and parameter counting	Information criteria
Temperature-dependent constraints	Linked temperature calculation
Parameter correlation (Pearson residual bootstrap)	Parameter correlation
Bootstrap SD and percentile confidence interval	Residual-bootstrap calculation
Automatic uncertainty display	Automatic display calculation
Salt dependence	Salt
Protonation dependence	Protonation
Buffer correction	Buffer Subtraction
Tandem reload mixing	What happens at a reload

Units are those shown by the application or export.

References

- FT-ITC Analysis repository
- FT-ITC Analysis website
- FT-ITC Project Viewer
- Software DOI: 10.5281/zenodo.14832177
- Malvern Instruments, *MicroCal ITC Analysis Software Using Origin User Manual*, MAN0577-02-EN-00 (20 May 2015), section 12.3.1, equations 2 and 4; section 12.3.6, equation 30 (dimer dissociation).
- **Help > Citation** for the current paper citation, versioned software citation, and BibTeX
- Philippe Dumas, *Isothermal titration calorimetry in the single-injection mode with imperfect mixing*, European Biophysics Journal 51, 77–84 (2022). The application's **Ideal continuous mixing** option is inspired by the paper but uses ideal mixing with a finite-injection numerical approximation; it does not implement the paper's imperfect-mixing model.
- Ernesto Freire, Arne Schön and Adrián Velazquez-Campoy, *Isothermal titration calorimetry: general formalism using binding polynomials*, Methods in Enzymology 455, 127–155 (2009). The application's **Discrete displacement** option uses this convention for concentrations and displaced-volume heat compensation, retaining FT-ITC's equilibrium solvers and offset convention.

Scientific claims and model-specific methods require the primary literature appropriate to the experiment and analysis.